

RANOPT AI

AI-Assisted Interactive 5G/6G RAN Decision Platform

System Design & Engineering Reference

Version 4.0

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CHAPTER 1

Product Description and Vision

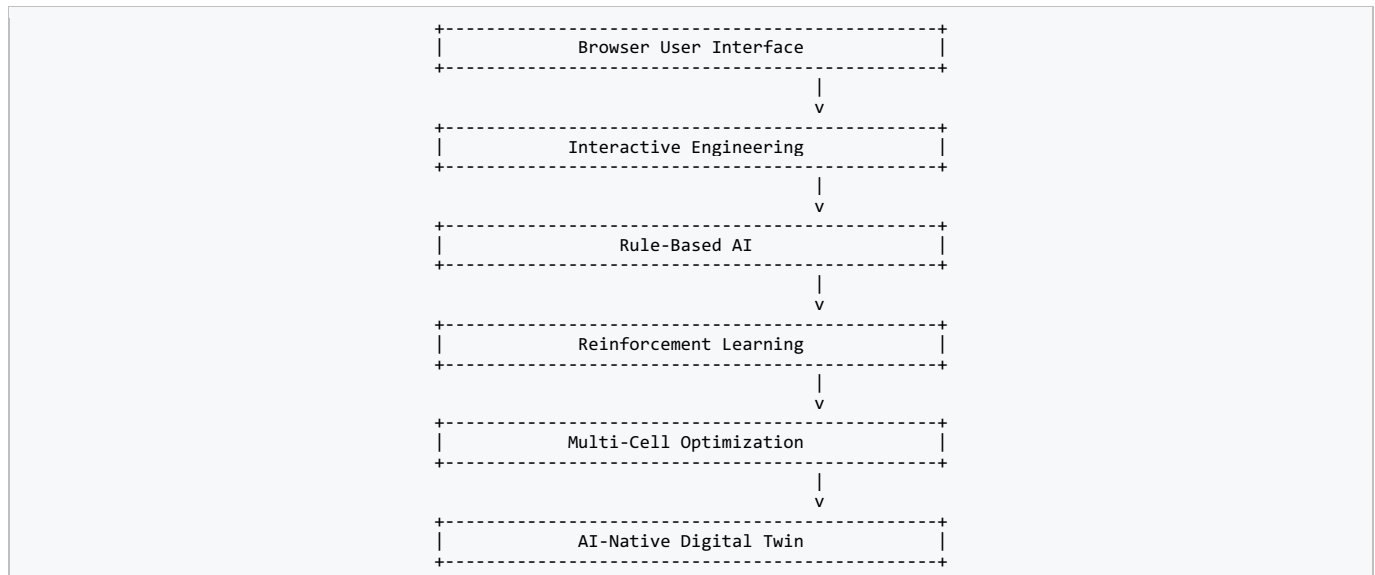
1.1 Motivation

The rapid evolution from fifth-generation (5G) mobile systems toward AI-native sixth-generation (6G) networks is significantly increasing the complexity of Radio Access Network (RAN) engineering. Modern cellular networks must simultaneously optimize coverage, capacity, spectral efficiency, energy consumption, latency, reliability, user mobility, and quality of experience across increasingly heterogeneous deployment environments. Emerging technologies such as massive MIMO, carrier aggregation, dynamic spectrum utilization, AI-native architectures, integrated sensing, and edge intelligence further expand the number of interacting design parameters that engineers must evaluate. As a result, traditional manual engineering approaches become progressively more challenging as network complexity increases.

Although sophisticated commercial planning and optimization tools are available, many engineering studies, technology evaluations, and “what-if” analyses still require significant expertise, lengthy configuration procedures, or large-scale simulation environments. Engineers frequently need a rapid and intuitive method for evaluating the impact of alternative radio configurations before committing to more detailed network planning or field validation.

Artificial Intelligence (AI), particularly Reinforcement Learning (RL), offers the potential to assist engineers by identifying effective operating points, recommending parameter adjustments, and learning from previous optimization outcomes. However, within RANOpt AI, AI is viewed as an **engineering decision-support assistant rather than a replacement for engineering judgement**. The platform preserves transparency by allowing users to understand, verify, and control every assumption, configuration, and optimization recommendation. Final engineering decisions always remain under the control of the user.

To address these challenges, **RANOpt AI** has been developed as an **AI (Reinforcement Learning)-Assisted Interactive 5G/6G RAN Decision Platform**. Its objective is to provide a browser-based environment that combines standards-aligned engineering models, interactive visualization, natural language and future voice interaction, and AI-assisted decision support to enable rapid evaluation of alternative radio configurations for an individual cell operating within a specified coverage area and deployment morphology. Rather than replacing existing planning and optimization tools, the platform complements them by providing a simple, interactive, and explainable environment for engineering analysis, technology evaluation, education, and design exploration.



1.2 Product Vision

Building on this motivation, the vision for RANOpt AI is to make advanced RAN engineering directly accessible to wireless engineers through an intuitive, browser-based workspace — evaluating, understanding, and optimising radio access network performance without replacing the planning and optimisation tools engineers already rely on.

By operating entirely within a standard web browser, the platform eliminates software installation and lets engineers rapidly modify network parameters, visualise performance changes, compare alternative scenarios, and run “what-if” analyses in real time.

Long term, RANOpt AI is intended to evolve into an area-wide Digital Twin Decision Platform for operational 5G Advanced and future 6G radio networks.

1.3 Product Positioning

RANOpt AI's positioning follows directly from this vision: it is a decision-support layer, not a replacement for established radio engineering systems.

Unlike conventional radio network planning tools, which are primarily designed for detailed network design and deployment, RANOpt AI emphasizes interactive engineering decision support — letting engineers quickly explore “what-if” scenarios, compare alternative configurations, evaluate AI-assisted optimization strategies, and visualize the impact of engineering choices through an accessible browser-based interface.

The platform is **not** intended to replace commercial radio planning packages, operational support systems (OSS), self-organizing network (SON) platforms, or real-time RAN controllers. Those systems perform functions such as detailed propagation prediction, large-scale network planning, live network management, automated optimization, and operational control. Instead, RANOpt AI serves as an engineering decision layer that complements these environments by enabling rapid design exploration, technology evaluation, AI demonstration, education, and standards-oriented performance assessment before more detailed planning or operational deployment is undertaken.

1.4 Browser-Based Interactive Engineering Platform

A core design objective is to make advanced 5G/6G RAN engineering accessible directly within a standard web browser — without dedicated software installation, complex runtime environments, or command-line execution — letting users begin engineering evaluations immediately from virtually any computing platform.

The browser-based architecture provides several practical advantages for engineering decision support. Network parameters can be modified interactively, simulation results are updated immediately, and graphical visualizations enable engineers to understand the relationships between system inputs and performance outputs without requiring scripting or complex command syntax. This interactive workflow encourages rapid exploration of alternative configurations, allowing users to perform “what-if” analyses efficiently during technology evaluation, network design studies, education, and collaborative technical discussions.

Unlike traditional command-line interfaces, which generally require users to remember commands, configuration files, or scripting procedures, the browser interface presents engineering parameters through intuitive controls, menus, graphical charts, and interactive visualisations, significantly reducing the learning curve for new users. Command-line environments remain valuable for large-scale automation, scripting, and batch simulation, which the browser interface is designed to complement rather than replace.

This browser architecture also provides a natural foundation for the conversational engineering capabilities described in Section 1.5.

Another important advantage of the browser-based implementation is portability. The platform can execute consistently across Windows, macOS, Linux, and other devices supporting modern web browsers, simplifying deployment for industry, research, education, and demonstrations. This architecture also supports incremental feature enhancement without requiring users to install or maintain dedicated software packages.

1.5 Conversational Engineering: Natural Language and Voice Interaction

One of the distinguishing characteristics of RANOpt AI is its vision of **conversational engineering**, where engineers interact with the platform using natural language in addition to conventional graphical controls. Rather than relying solely on menus, parameter forms, or command syntax, users can express engineering objectives using everyday technical language. This interaction model allows engineers to focus on solving wireless engineering problems instead of navigating software interfaces.

The platform has therefore been designed to support multiple complementary modes of interaction within the same browser-based environment. Users may configure simulations through conventional graphical controls, typed natural language commands, or, in future releases, voice interaction. Each interaction method ultimately modifies the same engineering model, ensuring that users retain complete visibility and control over every parameter and simulation assumption. Recent research into AI-native 6G management similarly explores conversational interfaces that allow operators to express intent in natural language while AI systems interpret and execute engineering tasks.

Typical engineering interactions may include requests such as:

- *“Increase the mid-band carrier bandwidth to 100 MHz.”*
- *“Compare AI On and AI Off performance.”*
- *“Optimize the scheduler for maximum cell throughput.”*
- *“Show the impact of higher user loading.”*
- *“Display the effect of carrier aggregation on cell-edge throughput.”*

Instead of requiring users to remember command syntax or manually modify multiple parameters, the platform interprets the engineering intent, updates the appropriate configuration settings, executes the analysis, and presents both the modified inputs and resulting performance metrics through the existing graphical interface. This conversational workflow reduces engineering effort while maintaining complete transparency of the underlying assumptions.

The browser-based implementation provides an excellent foundation for future voice interaction because modern web browsers already support speech recognition, audio input, and speech synthesis without requiring specialized hardware or dedicated client software. As these capabilities mature, engineers will be able to conduct engineering studies using either spoken or typed technical commands while receiving immediate graphical and numerical feedback through the same interactive interface.

Although conversational interaction simplifies operation, it does not replace engineering judgement: every AI recommendation and result remains visible to the user before acceptance.

1.6 AI (Reinforcement Learning)-Assisted Decision Support

Artificial Intelligence forms the foundation of RANOpt AI's long-term vision: an engineering decision-support system that assists wireless engineers in evaluating alternative radio configurations and recommending optimisation strategies, always under the engineer's final authority.

The current implementation combines deterministic engineering models with rule-based AI: standards-aligned calculations estimate representative KPIs, and rule-based logic evaluates these results to recommend engineering actions with full visibility of the underlying reasoning.

The long-term evolution introduces Reinforcement Learning (RL) as the primary AI methodology, in which an agent learns — through repeated interaction with the simulated environment — which actions best balance throughput, coverage, spectral efficiency, fairness, latency, and energy efficiency. Section 2.7 describes this framework in detail.

Explainability remains essential throughout: recommendations are expected to include the supporting rationale, expected KPI impact, and confidence level.

1.7 Current Engineering Scope

A fundamental design principle of RANOpt AI is to define a **clear and well-bounded engineering scope**. By explicitly identifying what the platform models today—and what is reserved for future releases—the system remains technically rigorous, computationally efficient, and transparent to its users. This approach ensures that engineers understand both the capabilities and the limitations of the current implementation, thereby increasing confidence in the interpretation of simulation results.

The present implementation focuses on **an individual radio cell operating within a specified coverage area and selected deployment morphology**. The user selects representative network parameters such as frequency band, duplex mode, channel bandwidth, antenna configuration, propagation environment, user density, scheduler type, traffic loading, and AI options. Using these inputs, the platform estimates representative cell-level Key Performance Indicators (KPIs), including coverage, cell throughput, user throughput, spectral efficiency, resource utilization, and other performance metrics relevant to engineering evaluation.

The engineering models are based on **standards-aligned analytical and deterministic calculations** rather than full-scale event-driven network simulation. This enables rapid execution of “what-if” analyses while maintaining consistency and repeatability of results. Since the same inputs always produce the same outputs, engineers can confidently compare alternative configurations without the statistical variability associated with large Monte Carlo simulations.

Although active users are represented within the selected cell, their locations are treated statistically rather than as individually moving devices. Consequently, the current implementation does **not** explicitly model user trajectories, handover procedures, inter-cell mobility, traffic evolution over time, or coordination among neighbouring cells. These simplifications are intentional and allow the platform to remain responsive for interactive engineering studies while preserving meaningful insight into radio performance.

Similarly, the platform does not attempt to replace detailed radio network planning or operational management systems. Functions such as high-resolution propagation prediction, drive-test correlation, neighbour-cell optimisation, SON control, real-time RAN operation, or OSS integration remain outside the scope of the current implementation. Instead, RANOpt AI provides

a rapid engineering decision-support environment for evaluating representative radio configurations before more comprehensive planning or operational validation is undertaken.

1.8 Future Evolution: From Interactive Decision Platform to AI-Native Digital Twin

RANOpt AI has been deliberately designed as an evolutionary platform. Rather than attempting to implement every aspect of future 6G network intelligence in its initial release, the platform adopts a phased development strategy in which new capabilities are introduced incrementally while maintaining a stable, transparent, and technically rigorous engineering foundation. This evolutionary approach allows each generation of the platform to remain immediately useful for engineering decision support while providing a clear roadmap toward increasingly sophisticated AI-assisted capabilities.

The next stage of evolution will strengthen the platform's AI capabilities by introducing the Reinforcement Learning-assisted optimisation described in Sections 1.6 and 2.7.

Beyond AI-assisted optimisation, future releases will extend the engineering scope from a single representative cell to **multiple interacting cells** within a geographical service area. This expansion will introduce realistic inter-cell interference, coordinated scheduling, user mobility, traffic evolution, and network-wide resource management. Engineers will then be able to study optimisation problems that cannot be represented within an isolated cell model, while retaining the same interactive browser-based workflow.

The ultimate vision is the development of an **AI-native RAN Digital Twin**, where a high-fidelity virtual representation of a real wireless network continuously mirrors its operational state. Such a digital twin will enable engineers to evaluate software updates, AI algorithms, parameter changes, and optimisation strategies within a safe virtual environment before deployment to the live network. This concept is widely recognised as a key enabling technology for AI-native 6G networks, providing a foundation for predictive optimisation, continuous validation, and intelligent network management.

As the platform matures, additional capabilities may include real-time network data integration, explainable AI (XAI), advanced visualisation, conversational engineering through natural language and voice interaction, and seamless interoperability with commercial planning, optimisation, and operational support systems. These enhancements will progressively transform RANOpt AI from an engineering decision-support tool into a comprehensive environment supporting the entire lifecycle of future intelligent radio access networks.

1.9 Engineering Philosophy

RANOpt AI's engineering philosophy follows directly from the principles introduced throughout this chapter — interactive engineering rather than fully automated optimisation, and a deliberate balance between engineering accuracy, computational efficiency, and usability. Deterministic models provide consistent, repeatable results, letting engineers explore design alternatives quickly while retaining sufficient technical realism for technology evaluation, education, and standards work.

In summary, the engineering philosophy of RANOpt AI can be expressed through five guiding principles:

- **Engineer in control** — AI assists; the engineer makes the final decision.
- **Explainable AI** — Every recommendation should be transparent and technically justifiable.
- **Interactive decision support** — Rapid exploration of engineering alternatives through an intuitive browser-based interface.
- **Standards-aligned engineering** — Models are grounded in accepted wireless engineering principles and relevant 3GPP specifications where applicable.
- **Evolution toward AI-native Digital Twins** — The platform is designed to grow from a cell-centric decision-support system into an area-wide AI-assisted Digital Twin while preserving transparency and engineering trust.

1.10 Chapter Summary

This chapter defined RANOpt AI as an interactive 5G/6G engineering decision platform and explained the need for rapid, transparent evaluation of increasingly complex radio choices.

It positions RANOpt AI as a browser-based decision-support layer that complements, rather than replaces, existing engineering workflows.

The roadmap progresses from cell-level analysis to multi-cell modelling, operator-data integration, and an AI-native network digital twin.

The concepts introduced in this chapter provide the foundation for the remainder of this document. The following chapters describe the engineering models, wireless assumptions, AI architecture, system implementation, validation methodology, and future evolution of RANOpt AI in progressively greater technical detail.

CHAPTER 2

System Architecture and Overall Workflow

2.1 System Overview

RANOpt AI integrates browser interaction, deterministic wireless calculations, AI-assisted recommendations, KPI computation, and visualization in one modular workflow.

The present architecture processes user-selected radio and loading parameters for a representative cell and presents the resulting KPIs through comparative dashboards.

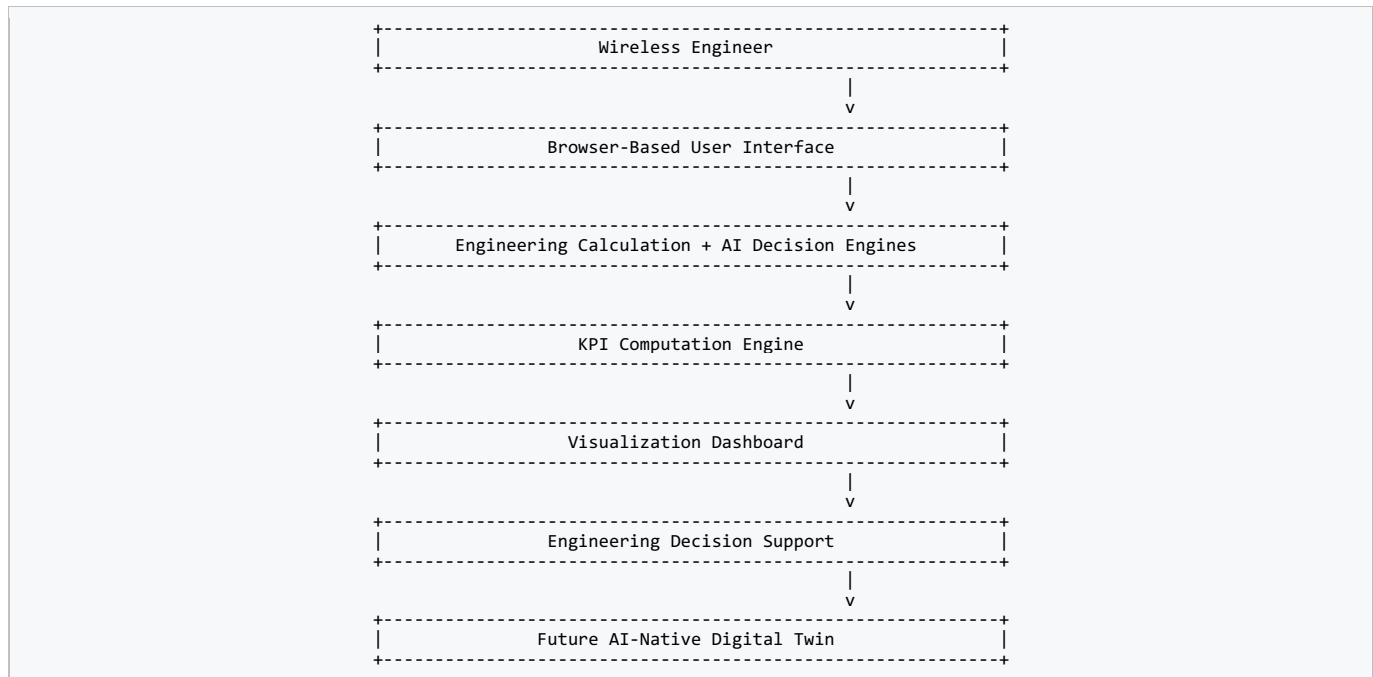
At the architectural level, RANOpt AI is organised into a set of cooperating functional subsystems, each responsible for a specific stage of the engineering workflow. These subsystems include:

- **Browser-Based User Interface** – Provides graphical interaction, parameter selection, future conversational (text/voice) engineering, and results presentation.
- **Engineering Calculation Engine** – Performs deterministic wireless engineering calculations based on standards-aligned analytical models.
- **AI Decision Engine** – Generates engineering recommendations using rule-based intelligence in the current implementation and explainable Reinforcement Learning (RL) in future releases.
- **KPI Computation Engine** – Calculates representative performance metrics including coverage, throughput, spectral efficiency, resource utilisation, latency estimates, and other engineering indicators.
- **Visualisation Engine** – Presents numerical results, engineering charts, comparative analyses, and AI recommendations through an interactive dashboard.
- **Future Digital Twin Framework** – Provides the architectural foundation for extending the platform toward multi-cell modelling, realistic mobility, inter-cell coordination, and AI-native network optimisation.

These subsystems operate as an integrated engineering workflow. User inputs initiate deterministic engineering calculations, AI analyses the resulting network conditions and recommends optimisation opportunities, KPIs are computed, and the results are presented through interactive visualisations that support engineering interpretation and decision making. Throughout this process, the engineer remains in complete control of every configuration parameter and every optimisation decision.

Figure 2.1 illustrates the high-level architecture of the RANOpt AI platform and the interaction among its principal functional modules.

Figure 2.1 – High-Level Architecture of RANOpt AI



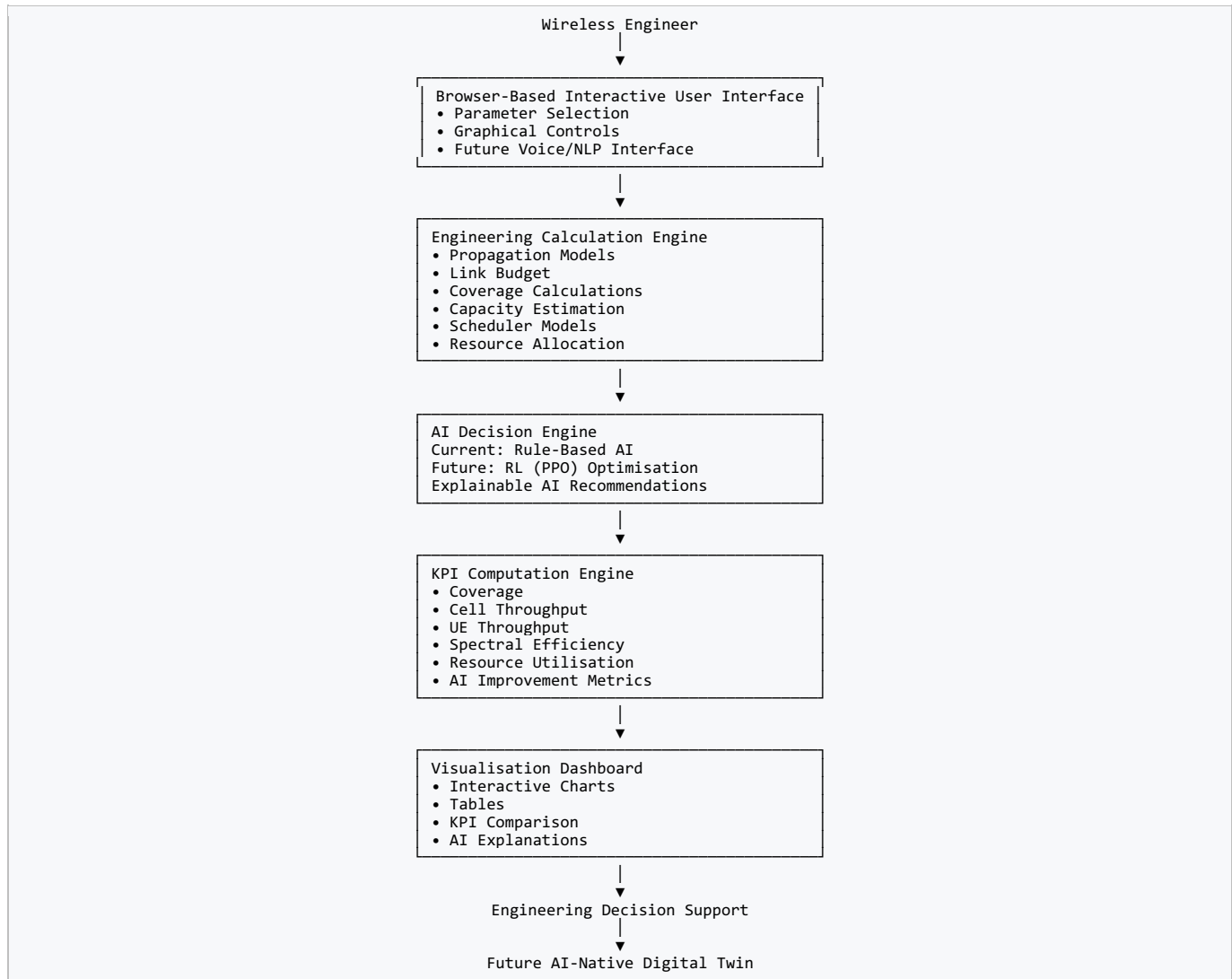
2.2 Overall System Architecture

The overall architecture of RANOpt AI has been developed using a **modular and layered design philosophy**, in which each functional subsystem performs a well-defined engineering task while interacting seamlessly with the other components. Such an architectural approach improves scalability, simplifies maintenance, and allows future capabilities to be incorporated without fundamentally changing the overall system structure. Modern system architecture practice recommends decomposing complex systems into cooperating functional modules with clearly defined responsibilities and interfaces.

User inputs pass through the calculation, AI, KPI, and visualization modules in a traceable sequence, allowing every displayed result to be related to its engineering assumptions.

The major architectural components of RANOpt AI are illustrated in **Figure 2.2**.

Figure 2.2 – Overall RANOpt AI System Architecture



The **Browser-Based User Interface** provides the primary point of interaction between the engineer and the platform. Engineers configure radio parameters, select deployment scenarios, execute engineering studies, and review results through an intuitive graphical interface. Future versions will extend this interface with natural language and voice interaction while preserving complete visibility of all engineering parameters.

The **Engineering Calculation Engine** forms the computational core of the platform. It implements deterministic, standards-aligned wireless engineering models covering radio propagation, link budget analysis, coverage estimation, capacity calculations, scheduler behaviour, and resource allocation. These calculations provide the quantitative foundation upon which subsequent AI analysis is performed.

The **AI Decision Engine** evaluates the calculated network state and recommends engineering improvements. The current implementation employs rule-based expert knowledge to generate explainable recommendations, while future releases will

incorporate **Reinforcement Learning (RL)** using the **Proximal Policy Optimisation (PPO)** algorithm. The AI engine is designed to assist engineers by identifying optimisation opportunities rather than making autonomous operational decisions.

The **KPI Computation Engine** derives representative network performance indicators from the engineering calculations. These include coverage probability, cell throughput, user throughput, spectral efficiency, resource utilisation, latency estimates, and AI-assisted performance improvements. The KPIs provide a quantitative basis for comparing alternative network configurations.

The **Visualisation Dashboard** transforms engineering results into interactive charts, tables, and graphical summaries that facilitate rapid interpretation of network behaviour. Comparative displays enable engineers to evaluate multiple scenarios, assess AI recommendations, and understand engineering trade-offs without requiring detailed examination of intermediate calculations.

Although the current implementation is intentionally **cell-centric**, the architecture has been designed to accommodate future expansion. The modular structure allows new capabilities—including multi-cell modelling, realistic user mobility, coordinated scheduling, inter-cell interference analysis, live network data integration, and Digital Twin functionality—to be incorporated with minimal modification to the existing system. This extensible architecture provides a practical migration path from today's interactive engineering platform to a comprehensive **AI-native 6G Digital Twin**.

2.3 Engineering Workflow

The engineering workflow of RANOpt AI has been designed to provide a simple, logical, and interactive sequence through which wireless engineers can configure network parameters, evaluate alternative scenarios, analyse system performance, and make informed optimisation decisions. Unlike conventional engineering software that often requires multiple independent tools or complex simulation procedures, RANOpt AI integrates the entire workflow within a single browser-based environment. This end-to-end approach reduces engineering effort while maintaining transparency, repeatability, and explainability throughout the analysis process. Structured engineering workflows are a fundamental principle of modern systems engineering because they improve traceability, consistency, and decision support across the design lifecycle.

The workflow begins with the engineer defining the radio network scenario. The user selects the deployment morphology, spectrum configuration, duplex mode, channel bandwidth, antenna configuration, scheduler type, traffic loading, AI options, and other radio parameters through the browser-based graphical interface. These inputs collectively define the engineering scenario to be analysed.

Once the configuration is complete, the Engineering Calculation Engine evaluates the scenario and passes the calculated network state to the AI and KPI modules.

The calculated network state is then passed to the AI Decision Engine. In the current implementation, the AI engine applies rule-based engineering knowledge to evaluate system performance and recommend potential optimisation opportunities. Future releases will replace or augment this capability with Reinforcement Learning (RL)-based decision support, enabling the platform to learn increasingly effective optimisation strategies from accumulated simulation experience while maintaining explainability of every recommendation.

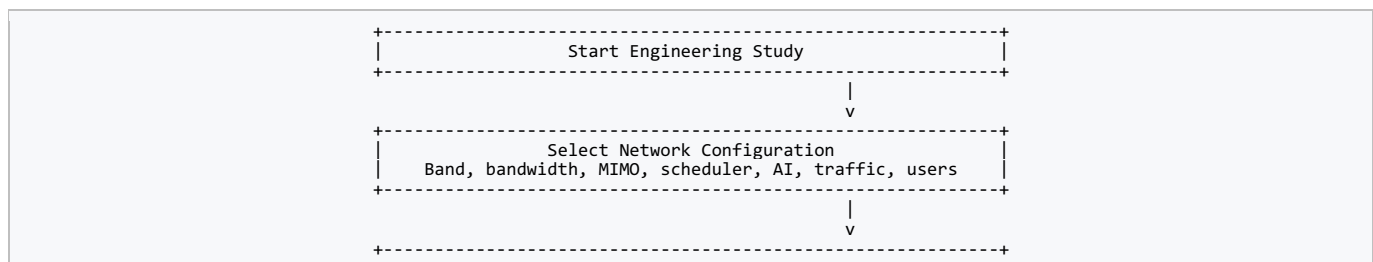
The KPI Computation Engine converts the calculated network state into the common comparison set used by the dashboard.

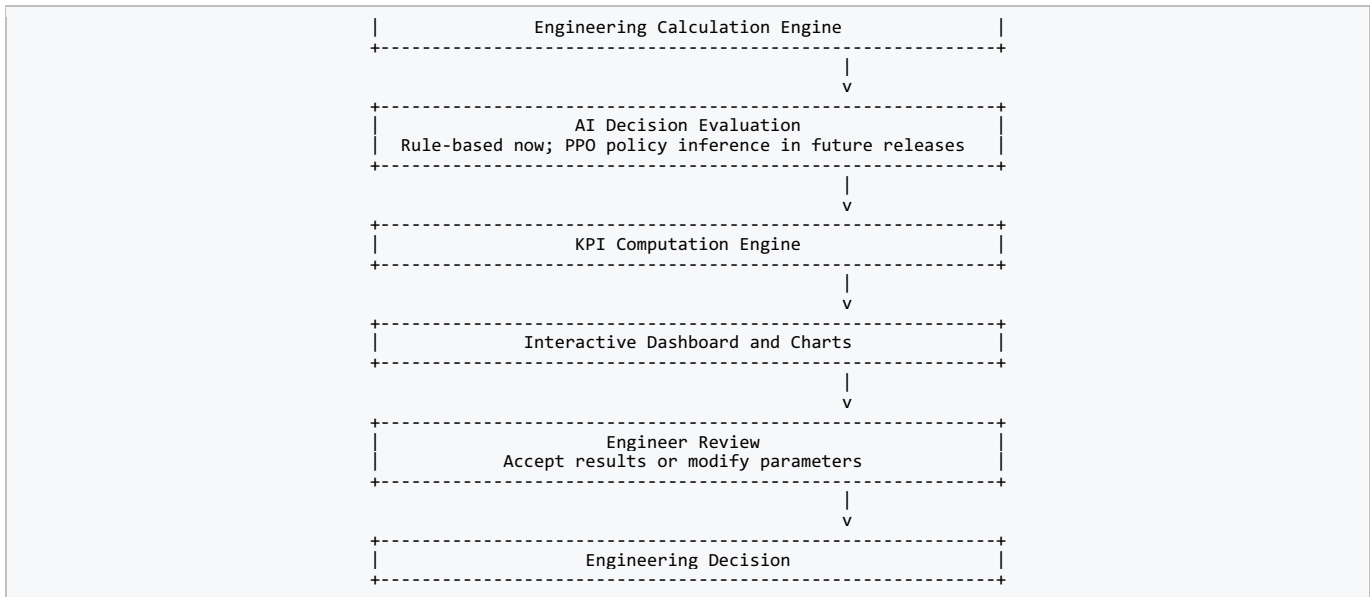
Finally, the Visualisation Dashboard presents the calculated results using interactive charts, tables, comparative graphs, and AI-generated recommendations. Engineers can immediately observe how modifications to network parameters influence overall system performance, compare multiple engineering scenarios, and identify the most appropriate configuration for the selected deployment conditions.

An important characteristic of the RANOpt AI workflow is its **iterative nature**. Rather than executing a single simulation, engineers are encouraged to repeatedly adjust parameters, evaluate new scenarios, compare results, and progressively refine network configurations. This interactive feedback loop supports rapid engineering exploration and facilitates a deeper understanding of complex relationships among radio parameters and network performance.

Figure 2.3 illustrates the complete engineering workflow implemented within RANOpt AI.

Figure 2.3 – RANOpt AI Engineering Workflow





The workflow closes with user review: the engineer can accept the result, modify assumptions, or run another comparison.

2.4 Browser-Based User Interface

The Browser-Based User Interface (UI) forms the primary interaction layer between the wireless engineer and the RANOpt AI platform. It has been designed to provide a simple, intuitive, and highly interactive engineering environment that enables users to configure radio network scenarios, execute engineering analyses, visualize system performance, and evaluate AI-assisted recommendations without requiring specialized software installation. By operating entirely within a modern web browser, the platform provides a consistent user experience across different operating systems while maintaining accessibility, portability, and ease of deployment. Modern web engineering practice recognizes browser-based applications as an effective approach for delivering complex engineering software with broad accessibility and maintainability.

Unlike traditional engineering tools that often require multiple configuration windows or command-line interfaces, the RANOpt AI interface is organized into a logical engineering workflow. Network configuration, AI options, simulation controls, and performance visualizations are presented within a single integrated workspace, allowing engineers to progress naturally from scenario definition to performance evaluation without changing applications or switching contexts.

The interface has been designed around the principle of **interactive engineering**. Engineers can modify parameters such as deployment morphology, frequency band, duplex mode, channel bandwidth, antenna configuration, scheduler selection, traffic loading, user density, AI settings, and propagation assumptions through graphical controls. Each modification immediately updates the engineering calculations, enabling rapid comparison of alternative configurations and efficient exploration of “what-if” scenarios.

To accommodate users with different levels of technical expertise, the interface supports two complementary operating modes:

Normal Mode

Normal Mode is intended for most engineering studies, demonstrations, education, and technology evaluation. Frequently used engineering parameters are presented through a simplified interface that minimizes unnecessary complexity while providing access to the most important configuration options. Performance results are displayed using concise KPI dashboards and graphical summaries that enable rapid interpretation of network behaviour.

Typical users include:

- Network planning engineers
- System engineers
- Technology evaluators
- Students and researchers
- Product demonstrations
- Executive technical reviews

Expert Mode

Expert Mode provides access to the complete set of engineering parameters, advanced radio configuration options, detailed KPI calculations, validation tools, and diagnostic information. This mode is intended for experienced wireless engineers who require fine control over simulation assumptions and wish to investigate the influence of individual engineering parameters on network performance.

Additional capabilities available in Expert Mode include:

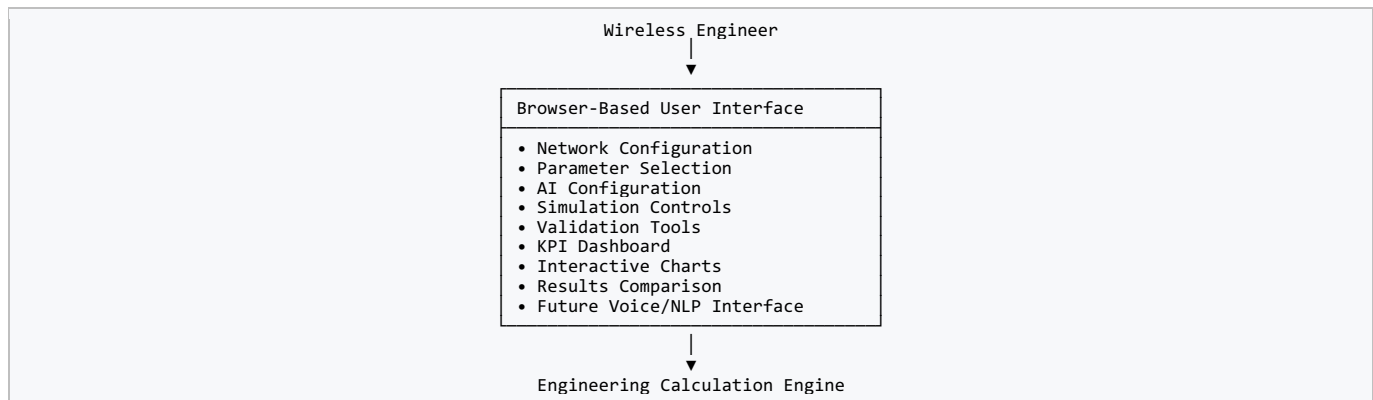
- Advanced radio configuration
- Detailed scheduler parameters
- AI configuration options
- Extended KPI reporting
- Validation and consistency checking
- Engineering diagnostics
- Advanced graphical visualisations

The browser interface also serves as the foundation for future **conversational engineering**. While the current implementation primarily relies on graphical interaction, the architecture has been designed to accommodate natural language and voice-based commands without altering the underlying engineering workflow. Engineers will eventually be able to configure simulations, request comparisons, and perform optimisation studies using conversational commands while continuing to observe all engineering parameters and intermediate results through the graphical interface.

The user interface has also been designed with **responsiveness and portability** as key objectives. It operates consistently on desktop and laptop computers using standards-compliant web technologies, allowing engineers to perform analyses without installing dedicated software. The modular interface architecture also simplifies future enhancements, enabling new engineering capabilities, AI functions, and visualisation tools to be incorporated without significant redesign. Responsive, browser-based interfaces are widely recognised as best practice for modern engineering web applications because they improve usability, portability, and long-term maintainability.

Figure 2.4 illustrates the principal functional elements of the RANOpt AI Browser-Based User Interface.

Figure 2.4 – Browser-Based User Interface Architecture



The Browser-Based User Interface therefore represents considerably more than a graphical front end. It functions as the central engineering workspace that integrates configuration, simulation control, AI-assisted optimisation, KPI visualisation, and engineering decision support into a unified interactive environment. Throughout every engineering session, the interface preserves the fundamental philosophy of RANOpt AI: **keeping the engineer in complete control while AI serves as an intelligent engineering assistant**.

2.5 Engineering Calculation Engine

The **Engineering Calculation Engine** is the computational core of the RANOpt AI platform. Its primary function is to transform user-selected network parameters into representative radio performance metrics using deterministic, standards-aligned wireless engineering models. Every coverage estimate, throughput prediction, spectral efficiency calculation, and capacity result presented by the platform originates from this subsystem. By separating engineering calculations from the user interface and AI modules, the architecture maintains modularity, transparency, and ease of future enhancement. This layered approach follows established systems engineering practice, where complex systems are decomposed into independent functional modules with clearly defined responsibilities.

Unlike full-scale event-driven network simulators that may require extensive computational resources and long execution times, the Engineering Calculation Engine employs analytical and deterministic models. Identical input parameters always

produce identical engineering results, enabling engineers to compare alternative network configurations objectively and repeatably. This deterministic behaviour is particularly valuable for engineering studies, technology evaluation, education, and AI-assisted optimisation because it eliminates statistical variability and allows the influence of individual parameters to be examined in isolation. Model-based wireless engineering remains an essential complement to AI-based techniques because engineering models provide physical insight, interpretability, and reproducibility.

The engine receives configuration parameters directly from the Browser-Based User Interface. Typical inputs include deployment morphology, operating frequency, duplex mode, channel bandwidth, carrier aggregation configuration, antenna characteristics, MIMO configuration, scheduler selection, user loading, propagation assumptions, and AI settings. These parameters define a representative radio environment that serves as the basis for all subsequent engineering calculations.

The calculations are organised into a sequence of logical processing stages that progressively build the radio performance model. Major computational functions include:

- Radio configuration validation
- Propagation loss estimation
- Link budget calculation
- Coverage prediction
- Signal-to-Interference-plus-Noise Ratio (SINR) estimation
- Channel Quality Indicator (CQI) determination
- Modulation and Coding Scheme (MCS) selection
- Spectral efficiency estimation
- Resource block allocation
- MAC scheduler performance estimation
- Cell throughput calculation
- User throughput estimation
- Coverage probability evaluation
- KPI generation

Each stage supplies the necessary inputs for subsequent calculations, ensuring a consistent engineering workflow while preserving complete traceability of intermediate results.

The propagation component implements representative large-scale radio propagation models appropriate for the selected deployment morphology. These models estimate path loss as a function of operating frequency, inter-site distance, antenna height, and environmental characteristics. The resulting received signal levels form the basis for subsequent link budget and SINR calculations. Future versions of the platform may incorporate additional propagation options and enhanced channel models as required for evolving 5G Advanced and 6G studies.

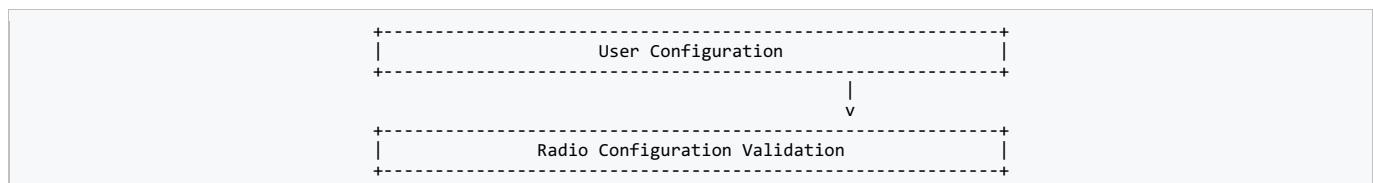
The link budget module combines transmitter power, antenna gains, feeder losses, receiver characteristics, propagation loss, and implementation margins to estimate received signal strength. These values are then used to calculate SINR, which determines the achievable modulation and coding scheme, spectral efficiency, and ultimately the user and cell throughput. The deterministic calculation sequence ensures that every reported KPI can be traced directly to the corresponding engineering assumptions and input parameters.

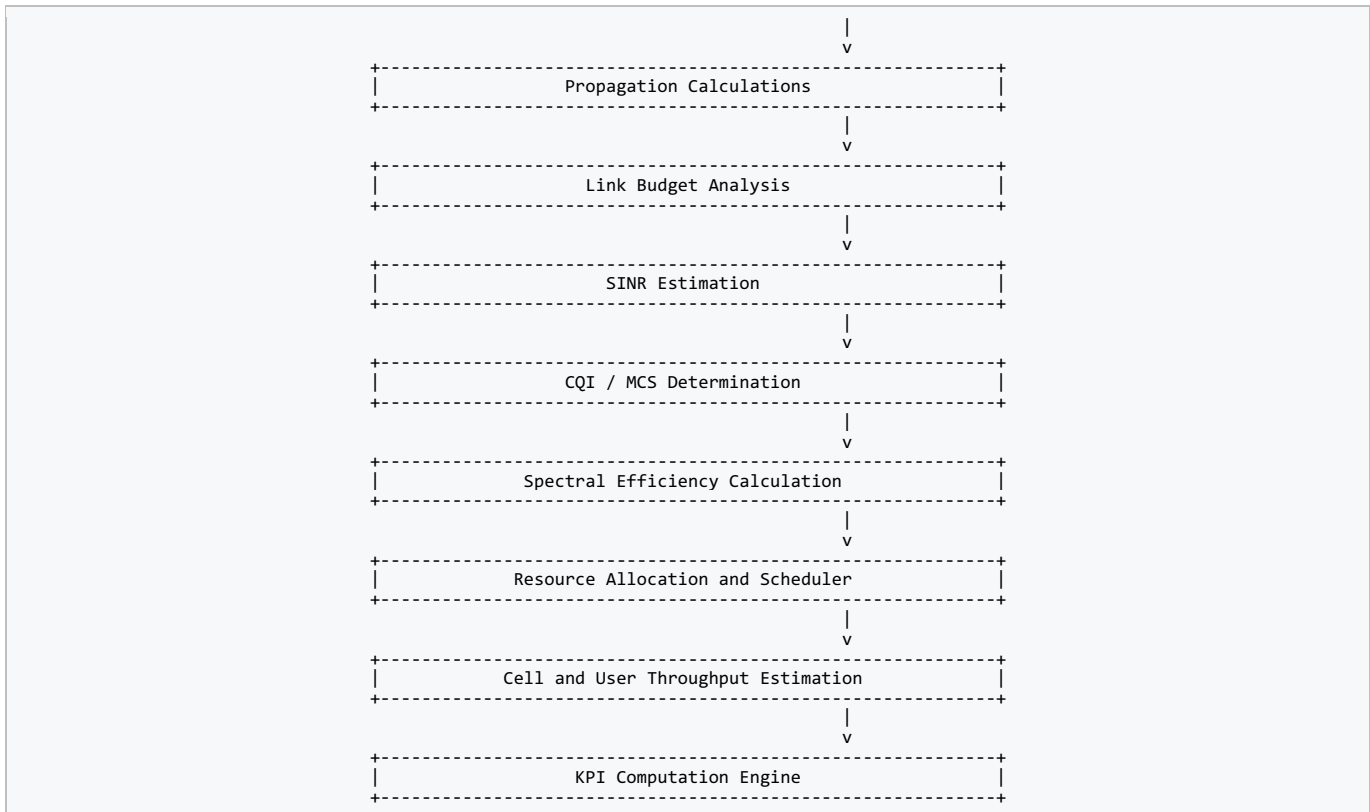
Resource allocation is subsequently evaluated using the selected MAC scheduling algorithm. The scheduler estimates how available radio resources are distributed among active users under the specified loading conditions. The resulting resource allocation determines representative user throughput, cell throughput, fairness, and resource utilisation. AI-assisted optimisation modifies selected scheduling or configuration parameters but does not alter the underlying engineering calculations themselves.

The outputs generated by the Engineering Calculation Engine are passed to the KPI Computation Engine, where representative network performance indicators are derived and subsequently displayed through the interactive visualisation dashboard. This separation between engineering calculations and KPI presentation simplifies system maintenance while allowing additional performance metrics to be incorporated in future releases without modifying the underlying computational framework.

Figure 2.5 illustrates the internal processing flow of the Engineering Calculation Engine.

Figure 2.5 – Engineering Calculation Engine





The Engineering Calculation Engine therefore provides the scientific foundation of RANOpt AI. By combining deterministic wireless engineering models, standards-aligned analytical calculations, and a modular architecture, it delivers accurate, repeatable, and explainable performance estimates while providing a stable computational platform upon which AI-assisted optimisation and future Digital Twin capabilities can be progressively built.

2.6 AI Decision Engine

The AI Decision Engine is the intelligent decision-support component of RANOpt AI. After the deterministic engineering calculations complete, it receives the resulting network state — coverage, SINR, CQI, spectral efficiency, scheduler utilisation, and other KPIs — and evaluates it for optimisation opportunities. Unlike autonomous control systems, it does not act independently: it provides explainable recommendations while the engineer retains the final decision, a human-in-the-loop design increasingly recognised as essential for trustworthy AI-assisted wireless networks.

Companion AI document: Detailed treatment of the rule-based decision logic, explainability method, PPO architecture, reward design, training data, inference, and AI validation is provided in RANOpt AI - AI Design & Implementation.

Current Rule-Based AI Implementation

The present implementation is a rule-based expert system: engineering knowledge is encoded as logical rules, derived from established wireless engineering practice, that evaluate network conditions and recommend corresponding actions — for example, increasing bandwidth when spectrum-limited, recommending higher-order MIMO where conditions allow, or advising carrier aggregation when component carriers are available. Because every recommendation traces to an explicit rule, this approach is fully explainable and auditable, which matters where transparency and traceability are essential.

Future Reinforcement Learning Implementation

Future versions will adopt Reinforcement Learning (RL) — specifically Proximal Policy Optimisation (PPO), chosen for its stability and sample efficiency — as the primary optimisation methodology. Rather than following predefined rules, an RL agent will learn optimisation strategies by repeatedly interacting with the engineering simulation: observing network state, selecting actions, and improving its policy from the resulting rewards. The full RL framework and PPO training methodology are described in Section 2.7 and in the companion AI document.

Explainable AI Decision Support

Whether a recommendation originates from rule-based logic or Reinforcement Learning, the AI Decision Engine is designed to explain itself: each recommendation is intended to show the network condition that triggered it, the suggested action, the

expected KPI improvement, and any trade-offs, together with a confidence level. This explainability is treated as essential for practical AI deployment in Open RAN and AI-native 6G networks, where operator trust and accountability are critical.

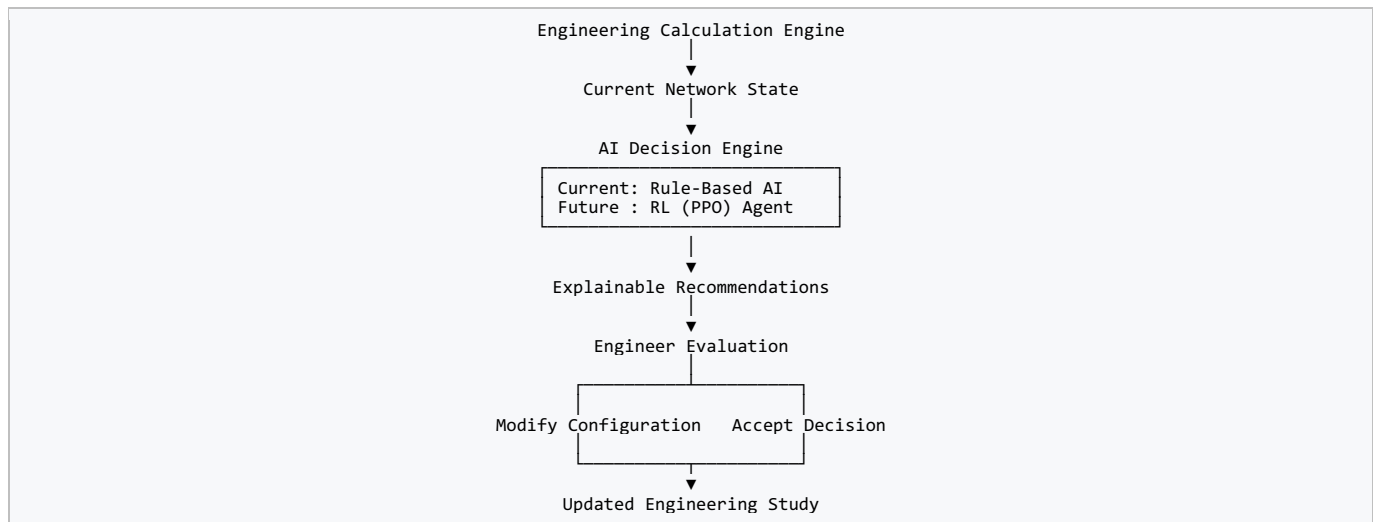
Human-in-the-Loop Operation

A fundamental design objective of RANOpt AI is that **AI never replaces the engineer**. The AI Decision Engine recommends optimisation strategies, but it does not automatically modify network parameters or execute engineering actions. The engineer reviews every recommendation, compares alternative scenarios, evaluates predicted KPI improvements, and determines whether the suggested optimisation should be adopted.

This collaborative workflow combines computational intelligence with professional engineering expertise: AI rapidly analyses complex interactions among multiple radio parameters, while the engineer contributes practical knowledge and operational judgement that cannot easily be reduced to an algorithm.

Figure 2.6 illustrates the functional operation of the AI Decision Engine.

Figure 2.6 – AI Decision Engine Architecture



The AI Decision Engine therefore provides the intelligence layer of RANOpt AI — combining deterministic engineering models, explainable rule-based reasoning today, and a future evolution toward Reinforcement Learning, while preserving the platform's central philosophy: AI augments engineering expertise, and the engineer remains in control.

2.7 Reinforcement Learning Framework

The platform architecture is designed to support Reinforcement Learning (RL) as its future primary optimisation methodology. RL is well suited to Radio Resource Management because wireless networks are dynamic environments in which engineering decisions continuously influence future performance; rather than following predefined rules, an RL agent learns effective strategies through repeated interaction with the environment.

Within RANOpt AI, the RL framework is implemented as a software agent that observes the current network state, selects an engineering action, receives a reward based on the resulting performance, and updates its policy — the standard Markov Decision Process (MDP) formulation of State, Action, Reward, and Policy.

Representative state variables include carrier configuration, MIMO mode, user loading, coverage, and SINR/CQI distribution; representative actions include scheduler parameter adjustment, carrier aggregation selection, MIMO mode optimisation, and resource-allocation strategy. The complete state and action specifications are defined in the companion AI document.

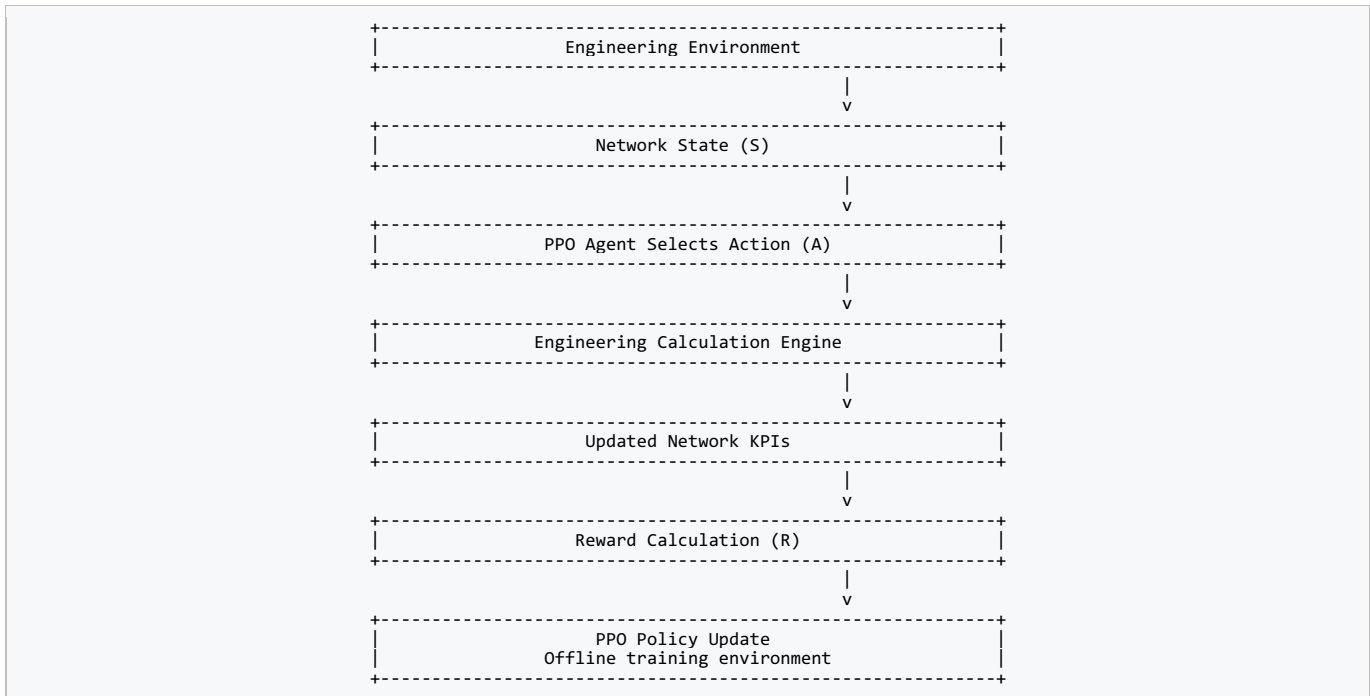
The reward function combines several KPIs — including throughput, cell-edge throughput, spectral efficiency, coverage, fairness, and latency — into a single multi-objective signal, so the agent learns to improve overall network behaviour rather than one metric at the expense of others.

The selected algorithm is Proximal Policy Optimisation (PPO), an actor-critic method chosen for its stable convergence and effective handling of large, complex decision spaces — properties well suited to radio resource management. Section 3.10 describes its application to MAC scheduling; full architecture and training details are in the companion AI document.

A key design feature is the separation of training from inference: PPO is trained offline, and the browser application performs inference only, loading a pre-trained policy to generate recommendations without incurring the cost of online training. This keeps the platform fast, deterministic, and fully browser-portable; Section 3.9.1 describes this two-stage architecture in detail.

Figure 2.7 illustrates the Reinforcement Learning framework adopted by RANOpt AI.

Figure 2.7 – Reinforcement Learning Framework



The Reinforcement Learning Framework thus represents the platform's future intelligence layer — evolving from rule-based assistance toward adaptive, AI-native decision support while preserving explainability and keeping the engineer in control of every decision.

2.8 Visualization and KPI Dashboard

The **Visualization and KPI Dashboard** provides the primary mechanism through which engineers interpret the results generated by RANOpt AI. While the Engineering Calculation Engine and AI Decision Engine perform the computational analysis, the dashboard transforms numerical outputs into intuitive graphical information that supports rapid engineering understanding and informed decision making. Effective engineering dashboards present not only performance metrics but also their engineering significance, enabling users to identify trends, compare scenarios, and understand the consequences of configuration changes.

The dashboard has been designed according to the philosophy that **engineering information should lead directly to engineering decisions**. Rather than presenting isolated numerical values, the dashboard organises related Key Performance Indicators (KPIs) into logical engineering categories, allowing users to understand how changes in radio configuration influence overall network behaviour.

The dashboard operates immediately after completion of each engineering study. Following the calculation of network performance and AI evaluation, representative KPIs are displayed through interactive charts, tables, summary indicators, and comparison plots. Engineers can immediately assess network behaviour without manually interpreting intermediate engineering calculations.

The current implementation presents representative radio performance indicators including:

- Coverage probability
- Cell throughput
- Average user throughput
- Cell-edge (5th percentile) user throughput
- Spectral efficiency
- Resource utilisation
- SINR distribution
- CQI distribution
- AI-assisted performance improvements
- Scheduler performance indicators

- Capacity utilisation

These KPIs collectively provide a comprehensive engineering assessment of the configured radio cell.

A distinguishing characteristic of the RANOpt AI dashboard is its ability to support **comparative engineering analysis**. Engineers can evaluate the effects of modifying network parameters by comparing multiple scenarios side-by-side. Typical comparisons include:

- AI Enabled versus AI Disabled
- Different deployment morphologies
- Alternative channel bandwidths
- MIMO configurations
- Carrier aggregation options
- Scheduler algorithms
- User loading conditions
- Coverage versus capacity trade-offs

Such comparisons enable engineers to understand not only whether network performance changes, but also **why** it changes.

The dashboard also serves as the primary interface for **Explainable AI (XAI)**. Whenever the AI Decision Engine generates an optimisation recommendation, the dashboard presents the associated engineering explanation alongside the predicted KPI improvements. Instead of displaying only the recommended action, the platform explains the engineering rationale, expected benefits, and potential trade-offs. This improves user confidence and supports informed engineering judgement. Explainable dashboards that combine AI recommendations with human-interpretable reasoning are increasingly regarded as an important element of trustworthy AI-assisted decision-support systems.

The browser implementation further enables highly interactive visualisation. Engineers can rapidly modify configuration parameters and immediately observe the resulting KPI changes without restarting the application or generating separate reports. This interactive feedback loop encourages exploratory engineering studies and significantly shortens the time required to evaluate alternative network configurations.

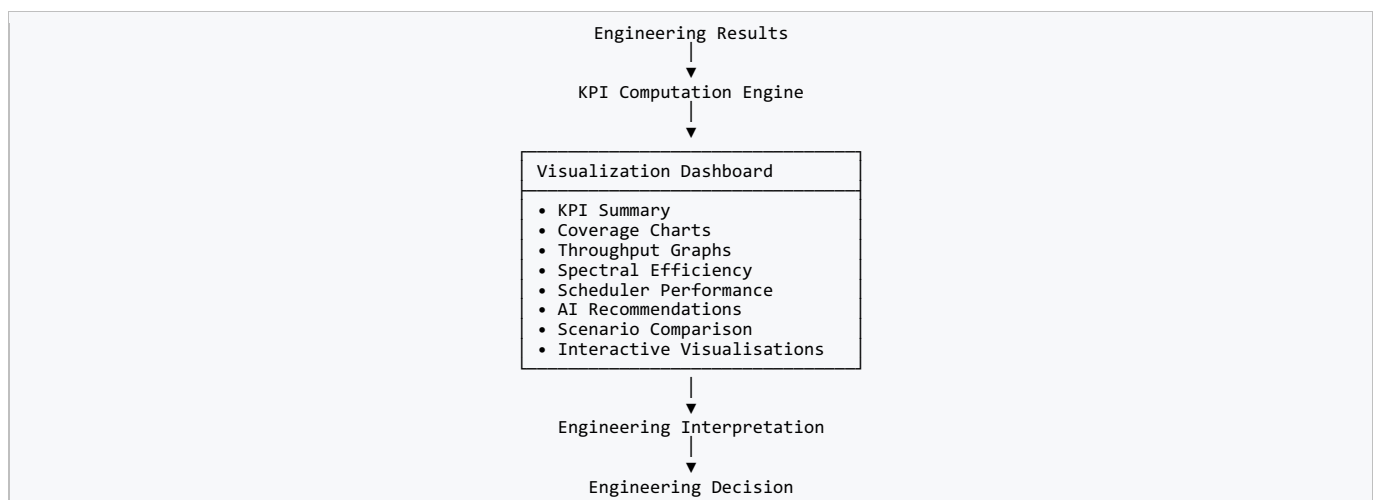
As the platform evolves toward an AI-native Digital Twin, the dashboard will also evolve beyond static KPI presentation. Future enhancements may include:

- Multi-cell network visualisation
- Geographic coverage maps
- User mobility visualisation
- AI confidence indicators
- Historical KPI trend analysis
- Predictive performance forecasting
- Digital Twin status monitoring
- Real-time network data visualisation

These capabilities will progressively transform the dashboard from a simulation results display into a comprehensive engineering decision-support environment.

Figure 2.8 illustrates the conceptual organisation of the Visualization and KPI Dashboard.

Figure 2.8 – Visualization and KPI Dashboard



The Visualization and KPI Dashboard therefore serves as the principal decision-support interface of RANOpt AI. By combining interactive visualisation, representative wireless KPIs, comparative analysis, and explainable AI recommendations within a unified browser-based environment, it enables engineers to rapidly interpret network behaviour, evaluate optimisation alternatives, and make informed engineering decisions while maintaining complete transparency of the underlying engineering calculations.

2.9 Future Digital Twin Architecture

The current implementation of RANOpt AI has been intentionally designed with an architecture that supports progressive evolution toward an **AI-native Network Digital Twin**. Although the present platform focuses on interactive engineering analysis of an individual radio cell, its modular design provides a natural migration path toward modelling, analysing, and optimising complete multi-cell wireless networks. This evolutionary strategy allows the existing engineering framework to be preserved while new capabilities are introduced incrementally, thereby protecting previous development effort and ensuring architectural continuity.

A **Network Digital Twin (NDT)** is a high-fidelity virtual representation of a physical communication network that continuously mirrors its operational characteristics. Unlike conventional simulation tools, which typically analyse predefined scenarios, a Digital Twin is intended to maintain a dynamic representation of the network by combining engineering models, operational data, artificial intelligence, and predictive analytics. Such environments enable engineers to evaluate proposed network changes, validate AI algorithms, and perform “what-if” analyses before modifications are introduced into the operational network. Network Digital Twins are increasingly regarded as a foundational technology for AI-native 6G systems because they enable safe experimentation, predictive optimisation, and continuous network evolution.

The Digital Twin architecture envisioned for RANOpt AI extends the current platform in several important dimensions. Rather than modelling a single representative cell, future versions will represent **multiple interacting cells** distributed across a geographical service area. Each cell will maintain its own radio configuration while interacting with neighbouring cells through interference, mobility, traffic distribution, and coordinated radio resource management.

The future architecture will progressively incorporate additional engineering capabilities, including:

- Multi-cell network topology
- Three-sector base station modelling
- Realistic user mobility
- Handover behaviour
- Inter-cell interference
- Dynamic traffic evolution
- Load balancing
- Coordinated scheduling
- Energy-aware optimisation
- AI-native network control

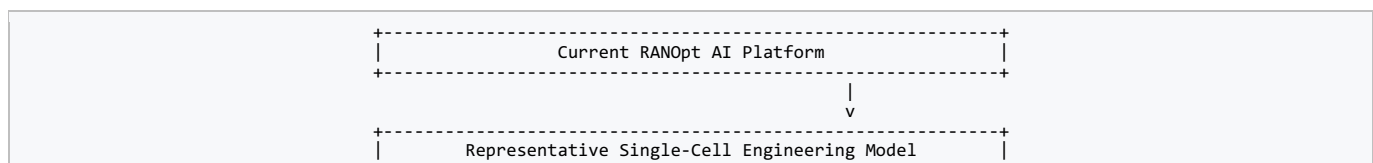
These enhancements will enable engineers to investigate optimisation problems that extend beyond the capabilities of a single-cell analytical model.

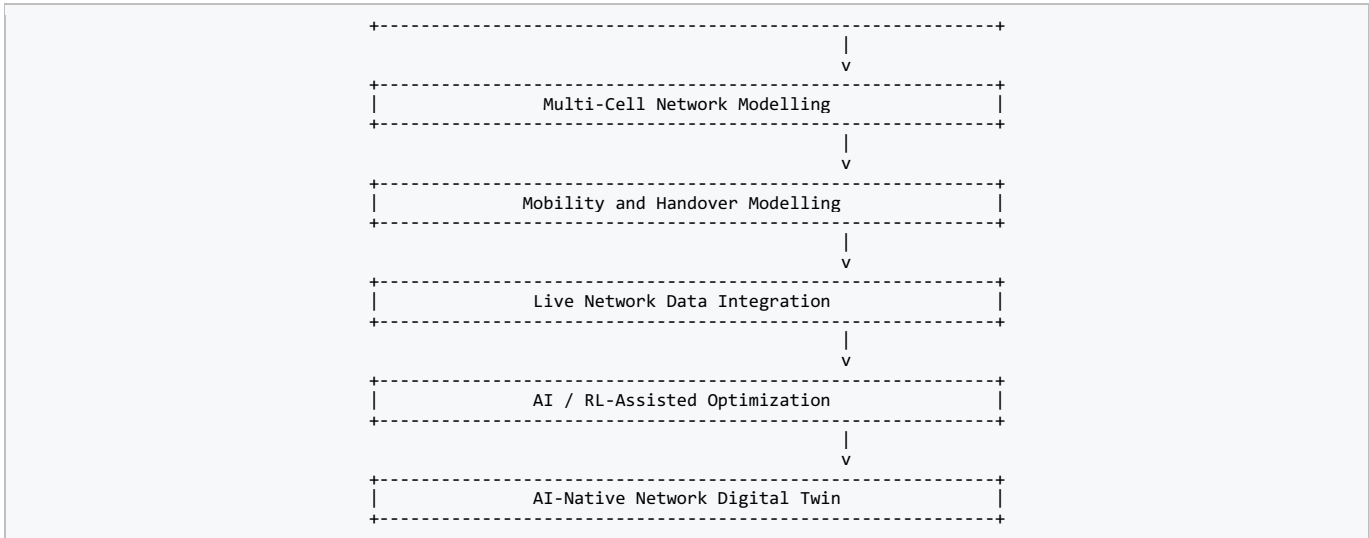
Artificial Intelligence will become an increasingly important component of the Digital Twin architecture. Rather than merely responding to existing network conditions, AI models will continuously analyse network behaviour, predict future operating conditions, evaluate alternative optimisation strategies, and recommend engineering actions before performance degradation occurs. Reinforcement Learning, predictive analytics, and Explainable AI (XAI) will operate together within the Digital Twin to support intelligent network management while maintaining complete engineering transparency. Recent AI-native 6G architectures similarly position Digital Twins as environments for predictive optimisation and safe validation of AI-driven decisions before deployment.

The proposed Digital Twin remains supervised: AI may analyse network behaviour and recommend actions, but acceptance and deployment remain engineering decisions.

Figure 2.9 illustrates the conceptual evolution of the RANOpt AI architecture toward an AI-native Digital Twin.

Figure 2.9 – Evolution Toward an AI-Native Digital Twin





The long-term architecture also provides opportunities for integration with external engineering environments. Future implementations may exchange information with network planning tools, Operations Support Systems (OSS), Open RAN management platforms, cloud-based AI services, and live network telemetry. Such interoperability would enable the Digital Twin to support not only engineering studies but also software validation, AI model evaluation, predictive maintenance, operational planning, and continuous network optimisation throughout the lifecycle of future 5G Advanced and 6G deployments.

2.10 Chapter Summary

This chapter described the modular architecture that connects browser interaction, deterministic engineering calculations, AI decision support, KPI computation, and visualization.

A distinguishing characteristic of the RANOpt AI architecture is the separation between **engineering calculations** and **AI-assisted decision support**. Wireless performance metrics are first derived using deterministic, standards-aligned engineering models. Artificial Intelligence then analyses the resulting network state and recommends optimisation strategies without altering the underlying engineering calculations. This separation preserves engineering transparency, repeatability, and explainability while allowing progressively more sophisticated AI algorithms to be incorporated in future releases.

The same modular structure provides a controlled path toward multi-cell modelling, mobility, operator-data integration, and a network digital twin.

CHAPTER 3

Wireless Engineering Models and Performance Analysis

3.1 Introduction

The accuracy and usefulness of any Radio Access Network (RAN) decision-support platform depend fundamentally on the engineering models that underpin its performance predictions. Regardless of the sophistication of the Artificial Intelligence (AI) algorithms employed, reliable optimisation can only be achieved when the underlying wireless engineering calculations accurately represent the behaviour of practical radio networks. For this reason, RANOpt AI adopts a **model-based engineering approach**, combining deterministic wireless communication models with AI-assisted optimisation to provide technically meaningful and repeatable engineering results. Modern systems engineering emphasizes that analytical models remain essential because they provide traceability, explainability, and physical insight into complex system behaviour.

Unlike large-scale network simulators that model thousands of individual users, network events, and protocol interactions, RANOpt AI focuses on representative **cell-level engineering analysis**. The objective is not to reproduce every protocol message or scheduling event, but to estimate the principal performance characteristics of a radio cell using standards-aligned engineering calculations. This approach enables rapid execution of interactive “what-if” studies while maintaining sufficient technical fidelity for engineering evaluation, technology comparison, AI development, and educational applications.

The engineering framework implemented in RANOpt AI follows the fundamental processing sequence used in practical wireless system design. Starting from user-selected network parameters, the platform successively evaluates propagation characteristics, link budget, received signal quality, spectral efficiency, resource allocation, scheduler performance, and finally the resulting Key Performance Indicators (KPIs). Each stage builds upon the results of the previous stage, producing a transparent calculation chain in which every performance prediction can be traced directly to the corresponding engineering assumptions.

The models implemented within the current platform are derived from well-established wireless communication principles and are aligned, where appropriate, with 3GPP radio access concepts used in 5G New Radio (NR). Representative engineering parameters include operating frequency, channel bandwidth, duplex mode, antenna configuration, carrier aggregation, propagation environment, user loading, scheduler selection, and radio resource management policies. These parameters collectively determine the achievable coverage, throughput, spectral efficiency, and capacity of the representative radio cell. The engineering emphasis is therefore placed on understanding the relationships among design parameters rather than reproducing every implementation-specific detail of a commercial network. 5G NR itself is built around these engineering concepts, including flexible spectrum usage, advanced antenna systems, and adaptable radio resource management.

An important characteristic of RANOpt AI is the use of **deterministic engineering models**. Given identical input parameters, the platform always produces identical engineering results. This repeatability enables engineers to evaluate alternative configurations objectively and understand the effect of individual design parameters without the statistical variability associated with Monte Carlo simulations. Such deterministic modelling also provides a stable environment for the future Reinforcement Learning (RL) framework, allowing AI algorithms to learn from consistent engineering behaviour while preserving explainability.

Although the current implementation intentionally focuses on a representative individual cell, the engineering models have been organised using a modular architecture that supports future expansion. Additional propagation models, advanced channel models, realistic mobility, multi-cell interference, Digital Twin functionality, and AI-native optimisation can be incorporated without changing the fundamental engineering workflow established in this chapter.

The remainder of this chapter describes the principal engineering models implemented within RANOpt AI. Beginning with radio propagation modelling and link budget analysis, the discussion proceeds through SINR estimation, spectral efficiency, scheduler behaviour, resource allocation, throughput calculations, and KPI generation. Together, these models provide the engineering foundation upon which the platform’s AI-assisted decision-support capability is built.

3.2 Radio Propagation Models

Radio wave propagation is one of the most fundamental aspects of wireless communication because it determines how transmitted electromagnetic energy travels from the base station to the user equipment (UE). Every performance metric calculated within RANOpt AI—including coverage, Signal-to-Interference-plus-Noise Ratio (SINR), throughput, spectral efficiency, and capacity—is ultimately influenced by the propagation model. Consequently, the selection of an appropriate propagation model is critical to obtaining realistic engineering results. Standardized propagation models such as those defined in **3GPP TR 38.901** provide a common framework for evaluating 5G NR systems across a wide range of deployment scenarios and frequencies.

Unlike site-specific ray-tracing techniques that require detailed geographical information and substantial computational resources, RANOpt AI employs **deterministic analytical propagation models** that represent the average large-scale behaviour of radio signals. This approach provides a practical balance between engineering accuracy and computational efficiency, enabling engineers to perform rapid interactive “what-if” analyses while maintaining consistency and repeatability of results.

The propagation calculations within RANOpt AI primarily estimate **large-scale path loss**, which represents the reduction in signal power as electromagnetic waves propagate through space and interact with the surrounding environment. Additional large-scale effects such as shadow fading may be incorporated through representative engineering assumptions where appropriate. Small-scale fading due to multipath propagation is represented separately through standardized channel models (CDL models), which are discussed later in this chapter.

The present implementation supports representative deployment morphologies commonly encountered in cellular networks, including:

- Dense Urban
- Urban
- Suburban
- Rural (future enhancement)

Each morphology is characterised by different propagation behaviour resulting from variations in building density, street geometry, vegetation, and terrain. These environmental characteristics influence the achievable coverage, received signal strength, and ultimately the radio performance experienced by users.

The propagation calculation uses several important engineering inputs selected by the user:

- Operating frequency
 - Cell type (macro or micro)
 - Inter-site distance (ISD)
 - Base station antenna height
 - User equipment antenna height
- Deployment morphology
 - Carrier bandwidth
 - Duplex mode

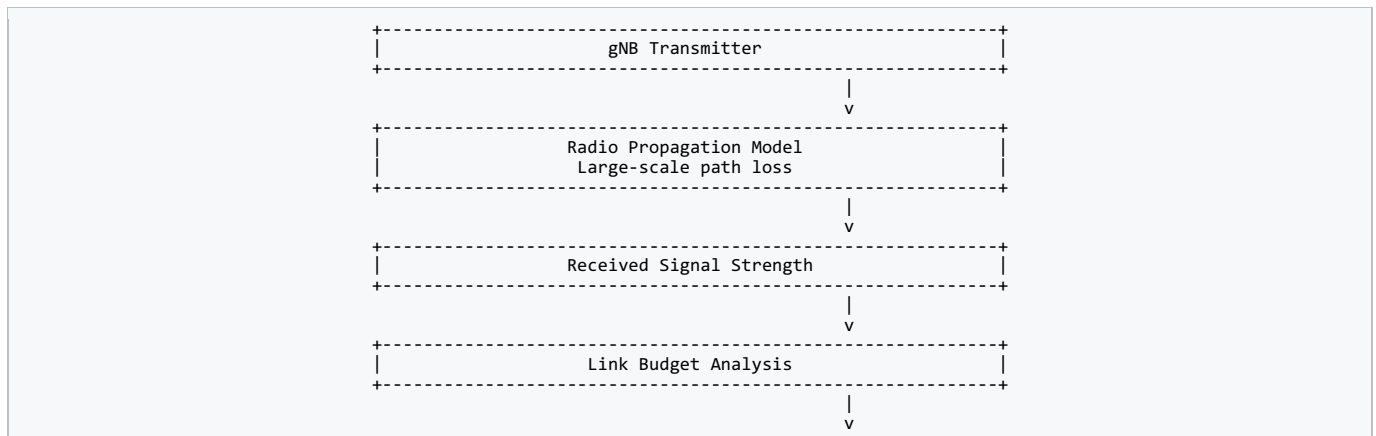
These parameters collectively determine the estimated path loss between the gNB and the UE.

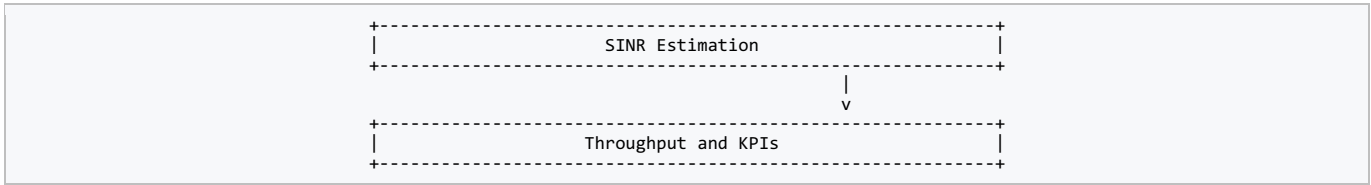
In accordance with the current architecture of RANOpt AI, propagation modelling is performed for **an individual representative cell** rather than an entire geographical network. This approach allows engineers to investigate how changes in frequency, antenna configuration, deployment environment, or cell geometry influence radio performance without the computational complexity associated with full multi-cell network simulation.

Although the current implementation focuses on large-scale propagation effects, the architecture has been designed to accommodate future enhancements. Planned developments include support for additional propagation scenarios, higher-frequency 6G Frequency Range 3 (FR3) bands, advanced suburban macrocell models introduced in later 3GPP releases, and tighter integration with the future AI-native Digital Twin architecture. 3GPP Release 19 is extending TR 38.901 with enhancements such as new suburban macrocell scenarios, improved antenna modelling, near-field propagation, and spatial non-stationarity to better support emerging 6G deployments.

Figure 3.1 illustrates the role of propagation modelling within the overall engineering calculation process.

Figure 3.1 – Role of Radio Propagation Modelling





The radio propagation model therefore provides the first major computational stage of the Engineering Calculation Engine. By estimating the average attenuation experienced by radio signals under representative deployment conditions, it establishes the physical foundation upon which all subsequent engineering calculations—including link budget, SINR, scheduler performance, throughput, and AI-assisted optimisation—are based

3.3 Link Budget Analysis

The **link budget** is one of the most important engineering calculations performed within RANOpt AI because it determines whether reliable communication can be established between the base station (gNB) and the User Equipment (UE). It provides a quantitative accounting of all gains and losses experienced by the transmitted radio signal as it propagates through the wireless channel. The resulting received signal power forms the basis for all subsequent engineering calculations, including Signal-to-Interference-plus-Noise Ratio (SINR), Channel Quality Indicator (CQI), spectral efficiency, throughput, and coverage. Consequently, link budget analysis is a fundamental step in the design and optimisation of modern cellular networks. 3GPP system-level evaluations rely on standardized propagation and channel models, such as those defined in **TR 38.901**, together with link-budget calculations to assess radio performance.

Within RANOpt AI, the link budget combines transmitter and receiver gains, propagation and implementation losses, and receiver characteristics to estimate signal strength at the UE.

The simplified downlink link budget implemented in RANOpt AI may be expressed as:

$$P_{RX} = P_{TX} + G_{TX} + G_{RX} - L_{Path} - L_{Misc}$$

where:

- P_{TX} = gNB transmit power
- G_{TX} = gNB antenna gain
- G_{RX} = UE antenna gain
- L_{Path} = propagation path loss
- L_{Misc} = miscellaneous losses including feeder losses, penetration losses, implementation margins, and other engineering allowances.

The calculated received power is then compared with the receiver sensitivity and thermal noise floor to determine whether reliable communication can be maintained under the selected deployment conditions.

The Engineering Calculation Engine considers a number of important parameters when computing the link budget, including:

- Operating frequency
- Channel bandwidth
- Duplex mode (FDD/TDD)
- gNB transmit power
- gNB antenna gain
- UE antenna gain
- gNB receiver noise figure (uplink)
- UE receiver noise figure (downlink)
- Deployment morphology
- Propagation model
- Cell radius or Inter-Site Distance (ISD)
- Implementation margins
- Cable and feeder losses (where applicable)

These parameters collectively determine the achievable received signal strength and therefore influence every higher-level performance metric calculated by the platform.

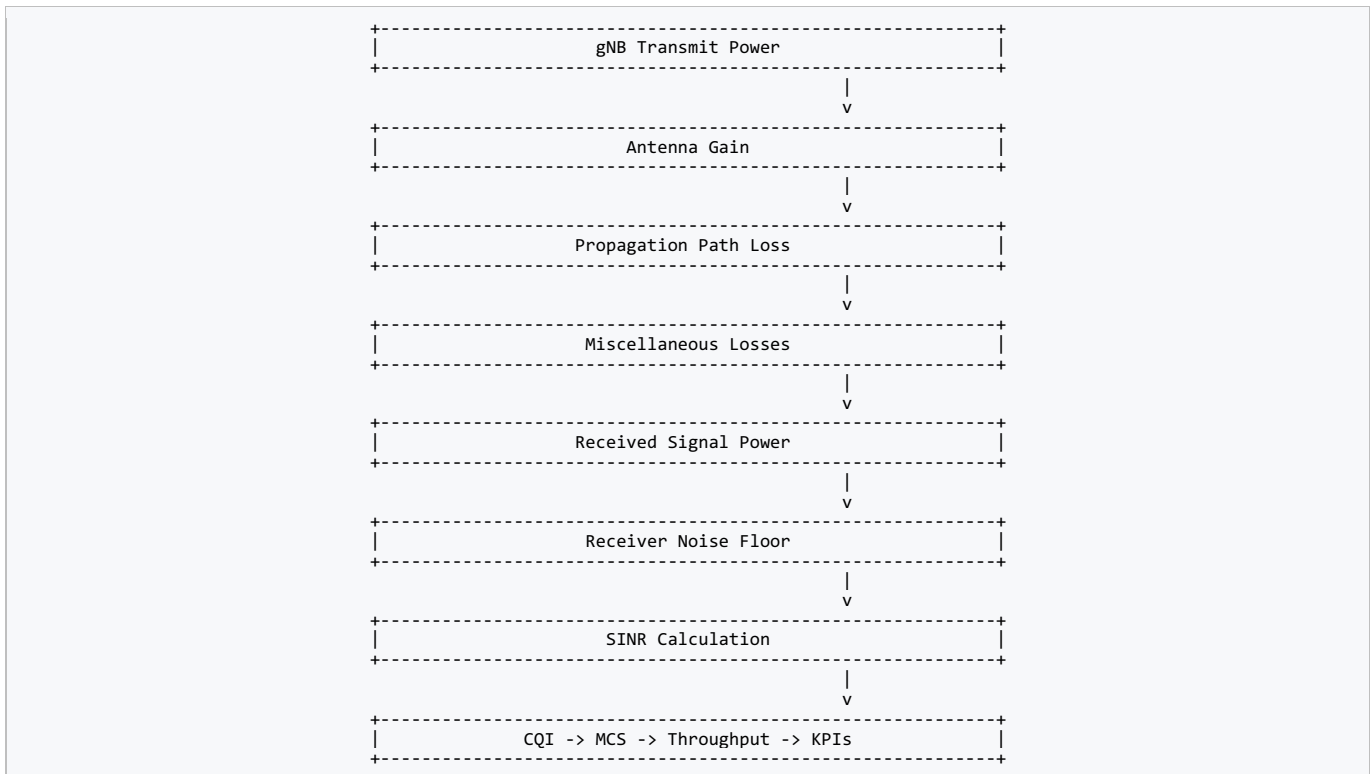
Unlike traditional link budget spreadsheets that provide only a single received power estimate, RANOpt AI integrates the link budget directly into the complete engineering workflow. The calculated received signal strength becomes the input to the SINR calculations, which subsequently determine CQI, Modulation and Coding Scheme (MCS), spectral efficiency, scheduler performance, and throughput. This integration enables engineers to understand how a change in any physical-layer parameter propagates through the remainder of the system and ultimately affects user experience.

The platform separately evaluates **downlink (DL)** and **uplink (UL)** link budgets because the two communication directions have different transmitter powers, receiver characteristics, antenna gains, and receiver noise figures. For example, the gNB typically transmits at significantly higher power than the UE, while the receiver noise figure differs between the UE receiver (downlink) and the gNB receiver (uplink). Accordingly, RANOpt AI allows independent modelling of these characteristics, providing more realistic engineering estimates of both coverage and capacity.

As wireless systems evolve toward 5G Advanced and 6G, additional factors such as Extremely Large Aperture Arrays (ELAA), higher-frequency FR3 operation, AI-assisted beam management, and dynamic antenna configurations will further influence link budget calculations. The modular architecture of RANOpt AI has therefore been designed to accommodate these enhancements without altering the fundamental engineering workflow. Emerging 3GPP Release 19 work extends channel modelling to support new propagation environments and advanced antenna characteristics relevant to future 6G systems.

Figure 3.2 illustrates the principal elements of the link budget calculation implemented within RANOpt AI.

Figure 3.2 – Link Budget Calculation



The link budget therefore forms the bridge between the physical propagation environment and higher-layer network performance. By accurately estimating received signal power under representative deployment conditions, it establishes the engineering foundation upon which SINR estimation, spectral efficiency modelling, scheduler behaviour, throughput prediction, and AI-assisted optimisation are subsequently built.

3.4 Noise Floor and SINR Calculations

Following the link budget analysis, the next major stage of the Engineering Calculation Engine is the determination of the **receiver noise floor** and the resulting **Signal-to-Interference-plus-Noise Ratio (SINR)**. SINR is one of the most influential performance parameters in a modern cellular network because it directly determines the modulation and coding scheme (MCS), spectral efficiency, achievable throughput, and ultimately the user experience. In practical wireless systems, received signal quality is limited not only by thermal noise but also by interference from neighbouring cells and other radio transmissions. Consequently, SINR provides a more realistic measure of radio link quality than Signal-to-Noise Ratio (SNR) alone.

3.4.1 Receiver Noise Floor

Every radio receiver generates a finite amount of internal noise that limits the weakest signal it can reliably detect. The receiver noise floor is determined by three principal components:

- Thermal noise
- Receiver bandwidth
- Receiver Noise Figure (NF)

The thermal noise power density at the standard reference temperature of 290 K is approximately:

-174 dBm/Hz

The receiver noise floor is then calculated as:

$$\text{Noise Floor (dBm)} = -174 + 10 \log_{10}(B) + NF$$

where:

- B = receiver bandwidth (Hz)
- NF = receiver Noise Figure (dB)

This relationship demonstrates two important engineering principles:

- Increasing receiver bandwidth increases the total integrated noise.
- Improving receiver Noise Figure lowers the receiver noise floor and improves receiver sensitivity.

Unlike many simplified educational tools, RANOpt AI models the **receiver Noise Figure independently for the UE and the gNB**, recognising that downlink and uplink receivers have different hardware characteristics.

For the **downlink**:

- UE receiver NF is used.

For the **uplink**:

- gNB receiver NF is used.

This distinction provides more realistic estimates of uplink and downlink coverage and throughput and reflects practical radio network implementation.

Signal-to-Interference-plus-Noise Ratio (SINR)

The platform then combines received signal power, interference, and receiver noise as follows.

SINR is defined as:

$$SINR = P_S / (P_I + P_N)$$

where:

- P_S = received signal power
- P_I = interference power
- P_N = receiver noise power

In decibel form:

$$SINR \text{ (dB)} = P_{RX} - 10 \log_{10}(P_I + P_N)$$

Unlike SNR, SINR incorporates the effects of both thermal noise and co-channel interference. In dense cellular deployments, interference frequently becomes the dominant limiting factor rather than receiver noise alone. Consequently, SINR is the preferred engineering metric for evaluating radio link quality in LTE, 5G NR, and future 6G systems.

3.4.2 Implementation within RANOpt AI

Within RANOpt AI, SINR is calculated after completion of the propagation and link budget analysis. The Engineering Calculation Engine combines:

- Received signal power
- Estimated interference
- Thermal noise
- Receiver Noise Figure

to obtain representative SINR values for the configured deployment scenario.

These SINR estimates subsequently drive:

- CQI determination
- Modulation and Coding Scheme (MCS) selection
- Spectral efficiency estimation
- Scheduler performance

- User throughput
- Cell throughput
- Coverage probability

Thus, SINR forms the direct bridge between the physical radio channel and higher-layer performance metrics.

3.4.3 Engineering Considerations

Several engineering parameters influence the calculated SINR, including:

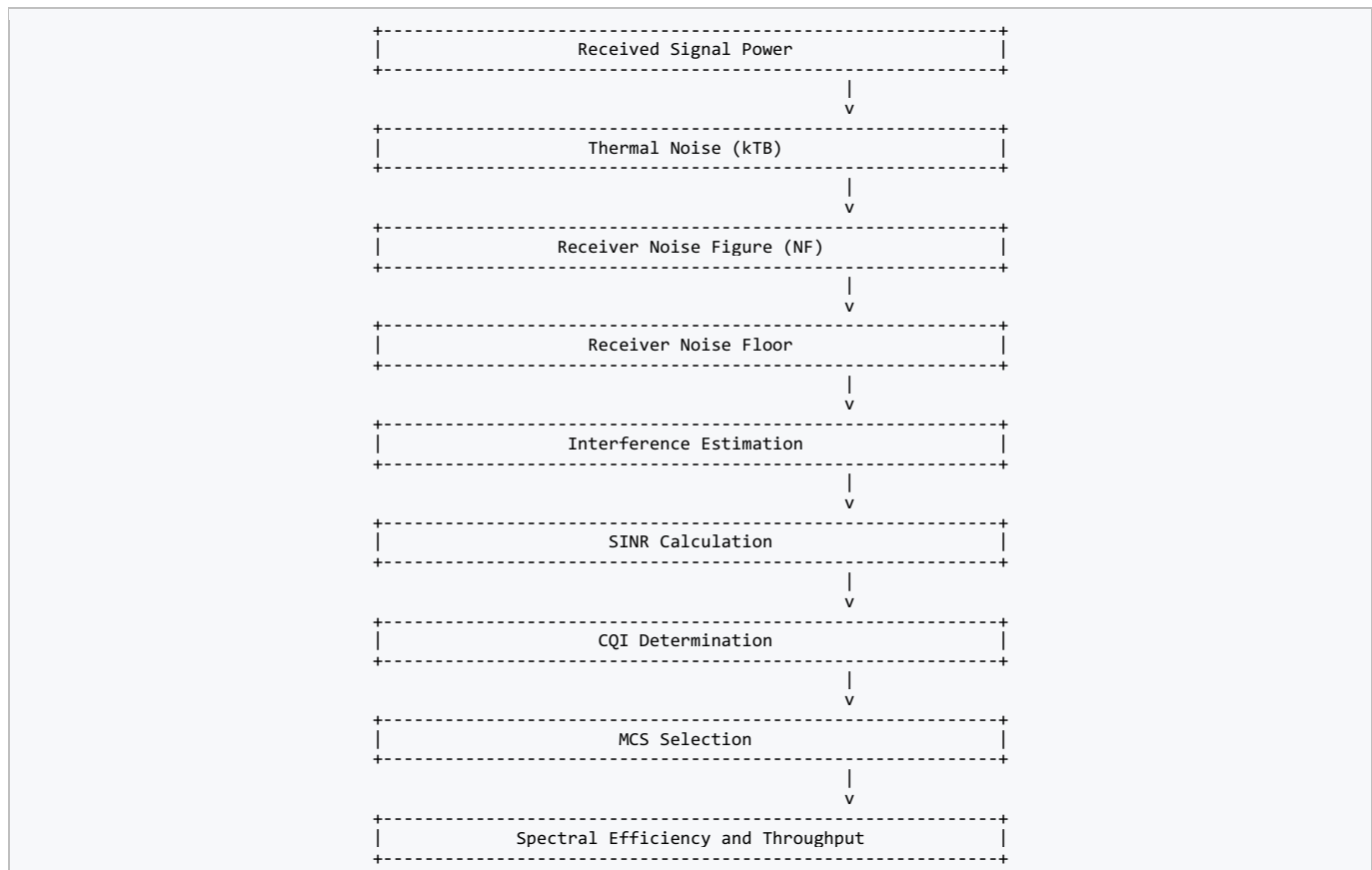
- Operating frequency
- Channel bandwidth
- Deployment morphology
 - Inter-site distance
 - Base station transmit power
 - Antenna gain
 - MIMO configuration
 - Carrier aggregation
 - User loading
 - Receiver Noise Figure
 - Interference conditions

By allowing engineers to modify these parameters interactively, RANOpt AI enables rapid investigation of the engineering trade-offs that determine achievable network performance.

3.4.4 Future Enhancements

The current implementation estimates representative interference using deterministic engineering assumptions suitable for a single-cell analytical model. As the platform evolves toward a multi-cell Digital Twin, future releases will incorporate explicit modelling of neighbouring-cell interference, coordinated scheduling, beamforming interactions, mobility-dependent interference dynamics, and AI-assisted interference management. These enhancements will provide increasingly realistic SINR estimation while preserving the modular architecture of the Engineering Calculation Engine.

Figure 3.3 – Noise Floor and SINR Calculation



The Noise Floor and SINR calculations therefore represent one of the most critical stages of the Engineering Calculation Engine. They establish the effective radio link quality upon which CQI, modulation and coding, scheduler behaviour, throughput estimation, and AI-assisted optimisation are based. By incorporating receiver bandwidth, independent UE and gNB Noise Figures, and representative interference conditions, RANOpt AI provides a technically realistic and standards-aligned foundation for evaluating 5G and future 6G radio performance.

3.5 Channel Models (CDL A–E)

While large-scale propagation models determine the average attenuation experienced by a radio signal, they do not describe the rapid variations caused by **multipath propagation**. In practical wireless environments, transmitted signals reach the receiver through multiple propagation paths created by reflections, diffraction, and scattering from buildings, vehicles, terrain, and other objects. The constructive and destructive combination of these multipath components produces small-scale fading, delay spread, and angular dispersion, all of which influence the achievable throughput and reliability of the radio link. To model these effects consistently, RANOpt AI adopts the **Clustered Delay Line (CDL)** channel models defined in **3GPP TR 38.901**. These standardized models are widely used for evaluating 5G NR physical-layer and system-level performance over frequencies from 0.5 GHz to 100 GHz.

Unlike deterministic propagation models, the CDL models statistically represent the radio channel by grouping multipath components into clusters, each characterised by its delay, power, angle of arrival, angle of departure, and polarization properties. This enables realistic representation of the wireless environment while maintaining computational efficiency suitable for engineering evaluation and system-level simulation.

The current implementation of RANOpt AI supports the five standardized CDL models defined by 3GPP:

3.5.1 Channel Model Simplification and Design Rationale

Although 3GPP TR 38.901 defines both Tapped Delay Line (TDL) and Clustered Delay Line (CDL) channel models, the current implementation intentionally (not being a physical layer simulator) does not generate complete multipath channel realizations. Specifically, it does not model Doppler effects, angle of arrival/departure, spatial clusters, polarization, beamforming channel matrices, or time-varying fading processes which are part of CDL. CDL uses the same basic delay profiles as TDL but

- Adds:
 - Azimuth AoA/AoD
 - Elevation AoA/AoD
 - Cluster spreads
 - Cross-polarization
 - Correlated antenna elements
 - Beamforming support
- Designed for:
 - Massive MIMO
 - Beamforming
 - Multi-user MIMO

Instead, the simulator combines large-scale propagation loss, user-selectable propagation environments, morphology-dependent path loss, antenna configuration, interference estimation, and calibrated spectral efficiency models to estimate achievable system performance. This level of abstraction is appropriate for system-level AI optimization, where the objective is to compare alternative network configurations and evaluate the impact of scheduler, spectrum, MIMO, AI, and deployment choices on overall network performance.

Accordingly, selecting CDL-A through CDL-E in the current implementation should be interpreted as representing different propagation environments with varying multipath richness rather than invoking a complete 3GPP CDL physical-layer channel model. Any corresponding reduction in spectral efficiency reflects the expected impact of the propagation environment on achievable performance and is intended as a calibrated system-level approximation rather than a direct simulation of clustered delay line channel behavior.

A full CDL implementation would require explicit generation of spatial MIMO channel matrices, angular statistics, beamforming, channel rank adaptation, CQI evolution, and time-varying fading. These capabilities are outside the scope of the present browser-based implementation but represent a natural enhancement for future versions of RANOpt AI as it evolves toward a high-fidelity digital twin for wireless network optimization.

Although all five models represent practical propagation environments, they differ primarily in their **delay spread**, **angular spread**, and the presence or absence of a dominant **Line-of-Sight (LOS)** component. According to 3GPP TR 38.901, **CDL-A**, **CDL-**

B, and CDL-C are Non-Line-of-Sight (NLOS) channel models, whereas CDL-D and CDL-E are Line-of-Sight (LOS) models with configurable K-factors.

Within RANOpt AI, the selected CDL model influences the effective radio channel quality through an **implementation efficiency factor**. Rather than explicitly simulating every multipath component, the platform uses representative efficiency adjustments that reflect the relative impact of different propagation conditions on achievable spectral efficiency and throughput. This analytical approach provides rapid execution while preserving the expected engineering trends associated with different channel environments.

From an engineering perspective:

- **CDL-D** and **CDL-E** generally provide the highest throughput because the dominant LOS component results in higher SINR, improved channel estimation, and more reliable MIMO operation.
- **CDL-A** represents relatively favourable NLOS propagation with modest multipath effects.
- **CDL-B** introduces additional multipath dispersion, reducing achievable spectral efficiency.
- **CDL-C** represents the most challenging NLOS environment among the five models and typically produces the greatest reduction in throughput due to increased delay spread and channel selectivity.

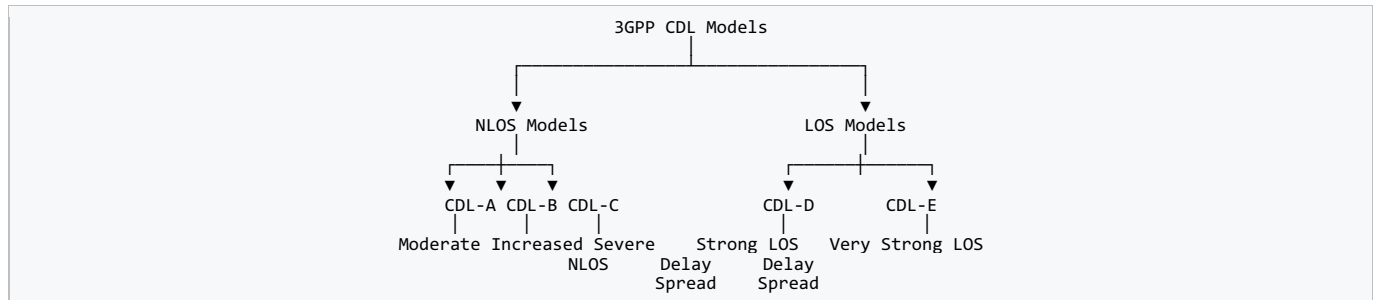
These differences are reflected within the Engineering Calculation Engine by appropriately adjusting the achievable spectral efficiency before throughput estimation. Consequently, the selected CDL model directly influences:

- Effective SINR
- Spectral efficiency
- Modulation and Coding Scheme (MCS)
- Cell throughput
- User throughput
- AI optimisation recommendations

An important design objective of RANOpt AI is to allow engineers to study how different radio environments influence network performance without requiring computationally intensive link-level simulations. By selecting different CDL models, users can rapidly compare representative LOS and NLOS deployment scenarios while maintaining deterministic and repeatable engineering results.

As 5G evolves toward **5G Advanced** and **6G**, channel modelling continues to develop. Recent 3GPP Release 19 work extends TR 38.901 to better represent emerging FR3 (7–24 GHz) deployments by introducing enhancements such as improved suburban macrocell modelling, realistic user-terminal antenna models, near-field propagation, spatial non-stationarity, and greater variability in cluster characteristics. These developments provide a foundation for future extensions of the RANOpt AI channel modelling framework.

Figure 3.4 – Representative CDL Channel Models



The CDL channel models therefore provide the bridge between large-scale propagation and realistic radio-link behaviour. By incorporating representative LOS and NLOS channel characteristics defined by 3GPP, RANOpt AI produces engineering results that more closely reflect practical wireless deployments while preserving the computational efficiency required for interactive AI-assisted engineering analysis.

3.6 Spectral Efficiency Modelling

Spectral efficiency is one of the most important performance measures used in wireless communication because it quantifies how effectively the available radio spectrum is utilized. It represents the amount of useful information that can be transmitted over a given bandwidth and is normally expressed in **bits per second per Hertz (bits/s/Hz)**. Since radio spectrum is both finite and expensive, improving spectral efficiency is a primary objective of modern cellular systems including LTE, 5G NR, and future 6G networks.

Within RANOpt AI, spectral efficiency serves as the principal link between physical-layer radio quality and higher-layer network performance. Once the propagation model, link budget, and SINR have been determined, the Engineering Calculation Engine estimates the achievable spectral efficiency for the selected deployment scenario. This value subsequently forms the basis for throughput, capacity, and resource utilisation calculations.

In its simplest form,

$$\eta = \text{Throughput} / \text{Channel Bandwidth}$$

where

- η = Spectral Efficiency (bits/s/Hz)
- Throughput = useful data rate (bits/s)
- Bandwidth = occupied channel bandwidth (Hz)

This relationship illustrates an important engineering concept: for a fixed bandwidth, increasing spectral efficiency directly increases achievable throughput, while for a required throughput, higher spectral efficiency reduces the spectrum required.

From information theory, the theoretical upper limit is given by the **Shannon Capacity Theorem**:

$$\eta_{\text{max}} = \log_2(1 + \text{SINR})$$

This equation represents an ideal upper bound under additive white Gaussian noise (AWGN) conditions. Practical wireless systems operate below this limit because of implementation losses, imperfect channel estimation, signalling overhead, scheduling constraints, and coding limitations.

In practical 5G NR systems, spectral efficiency depends upon several interacting engineering factors:

- Signal-to-Interference-plus-Noise Ratio (SINR)
- Modulation order (QPSK, 16-QAM, 64-QAM, 256-QAM)
- Coding rate
- Channel Quality Indicator (CQI)
- MIMO configuration
- Carrier Aggregation
- MAC scheduler performance
- Radio propagation conditions
- CDL channel model

Higher SINR allows the network to select higher-order modulation and higher coding rates, thereby increasing spectral efficiency. Conversely, poor radio conditions require more robust modulation and coding, reducing the achievable data rate but improving link reliability. This adaptive modulation and coding process is fundamental to 5G NR operation.

Within RANOpt AI, spectral efficiency is not calculated solely from Shannon capacity. Instead, the Engineering Calculation Engine combines deterministic engineering models with representative implementation factors that reflect practical network behaviour. These include:

- SINR-dependent modulation and coding selection.
- CDL channel efficiency factors.
- MIMO spatial multiplexing gain.
- Carrier aggregation gain.
- Scheduler efficiency.
- AI-assisted optimisation improvements.

This approach produces spectral efficiency estimates that are representative of practical deployments while remaining computationally efficient for interactive engineering studies.

Channel-dependent efficiency adjustments feed directly into spectral-efficiency estimation. Line-of-Sight environments represented by CDL-D and CDL-E generally permit higher efficiency than the more challenging Non-Line-of-Sight environments represented by CDL-B or CDL-C, without requiring computationally intensive physical-layer simulation.

Similarly, advanced radio technologies incorporated within the platform influence spectral efficiency:

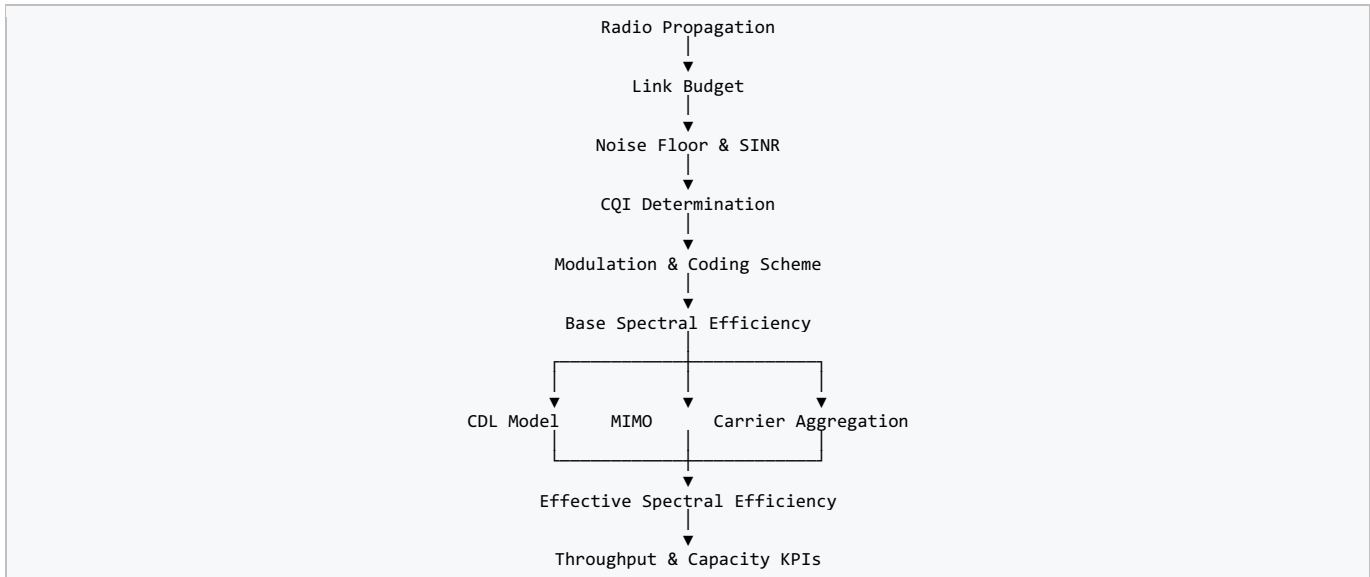
- Massive MIMO improves spatial multiplexing capability.
- Carrier Aggregation increases usable bandwidth while improving overall cell capacity.
- AI-assisted scheduler optimisation improves resource utilisation.
- Better CQI enables higher Modulation and Coding Schemes (MCS).
- Improved beamforming increases effective SINR.

The combined effect of these technologies is reflected in the final spectral efficiency used for throughput calculations.

As wireless systems evolve toward **5G Advanced** and **6G**, further improvements in spectral efficiency are expected through AI-native scheduling, Extremely Large Aperture Arrays (ELAA), advanced beam management, Integrated Sensing and

Communications (ISAC), semantic communications, and more efficient waveform designs. The modular architecture of RANOpt AI has therefore been designed so that future spectral efficiency enhancements can be incorporated without changing the overall engineering workflow.

Figure 3.5 – Spectral Efficiency Determination



The Spectral Efficiency Model therefore forms the critical connection between physical-layer radio quality and overall network performance. By integrating SINR, adaptive modulation and coding, channel conditions, MIMO, carrier aggregation, scheduler behaviour, and AI-assisted optimisation into a unified analytical framework, RANOpt AI provides representative and repeatable estimates of spectrum utilisation that closely reflect the behaviour of practical 5G NR systems while remaining suitable for rapid interactive engineering analysis.

3.7 MAC Scheduler Models

The **Medium Access Control (MAC) scheduler** is one of the most critical components of a cellular radio access network because it determines how the available radio resources are allocated among active users. Although the physical layer determines the theoretical transmission capability of the radio channel, it is the MAC scheduler that decides **which User Equipment (UE) is served, when it is served, and how many radio resources are assigned to it**. Consequently, scheduler performance has a direct impact on cell throughput, user throughput, latency, fairness, Quality of Service (QoS), and overall spectrum utilization. In 5G NR, the scheduler resides in the **gNB MAC layer** and independently schedules downlink and uplink transmissions. Importantly, **3GPP does not standardize the scheduling algorithm itself**, allowing equipment vendors to differentiate their products through proprietary scheduler implementations.

Within RANOpt AI, the MAC scheduler is modelled using deterministic engineering algorithms that reproduce the behaviour of practical cellular schedulers while maintaining rapid execution suitable for interactive engineering analysis. Rather than simulating every scheduling decision at slot level, the Engineering Calculation Engine estimates the average impact of scheduler behaviour on resource utilization and user performance.

The scheduler receives information from several components of the radio system, including:

- Channel Quality Indicator (CQI)
- Buffer status
- Quality of Service (QoS) requirements
- Active user count
- Available bandwidth
- Carrier aggregation configuration
- MIMO capability
- AI optimisation status

Using these inputs, the scheduler determines how Physical Resource Blocks (PRBs) are distributed among active users during each scheduling interval.

3.7.1 Proportional Fair Scheduler

The primary scheduler implemented within the current version of RANOpt AI is the **Proportional Fair (PF)** scheduler. The PF scheduler is widely adopted in LTE and 5G because it provides an effective balance between maximizing total cell throughput and maintaining fairness among users. Rather than always selecting the user with the best instantaneous radio conditions, the PF scheduler also considers the user's historical average throughput. Users with favourable channel conditions are therefore scheduled more frequently, but users experiencing temporarily poorer radio conditions continue to receive a reasonable share of resources. This balance between efficiency and fairness has made PF the de facto scheduler in many commercial systems.

The PF scheduling metric can be expressed as:

$$PF_i = R_i^{inst} / R_i^{avg}$$

where:

- R_i^{inst} = instantaneous achievable data rate determined from CQI and MCS.
- R_i^{avg} = historical average throughput of user i .

The scheduler assigns resources to the UE having the highest PF metric. Users with excellent instantaneous radio conditions receive higher priority, while users that have received fewer resources in previous scheduling intervals gradually gain priority through a lower historical average throughput. This adaptive process simultaneously improves cell throughput and fairness.

3.7.2 Scheduler Models Supported

Although the present implementation primarily employs the Proportional Fair scheduler, the modular architecture permits future implementation of additional scheduling strategies commonly used in wireless systems, including:

This extensible design ensures that future versions of RANOpt AI can evaluate different scheduler philosophies without requiring modifications to the remainder of the Engineering Calculation Engine.

3.7.3 Scheduler Influence on Network Performance

The MAC scheduler directly influences numerous performance indicators generated by RANOpt AI, including:

- Cell throughput
- Average UE throughput
- 5th percentile (cell-edge) throughput
- Spectral efficiency
- Resource utilization
- User fairness
- Latency
- Quality of Experience (QoE)

For example, a Maximum Rate scheduler may maximize overall cell throughput by preferentially serving users with excellent channel conditions, but this can significantly reduce performance for users near the cell edge. Conversely, a Round Robin scheduler distributes resources equally among users but generally sacrifices overall spectral efficiency. The Proportional Fair scheduler represents a practical compromise between these two extremes and is therefore well suited for representative engineering analysis.

3.7.4 AI-Assisted Scheduling

One of the distinguishing features of RANOpt AI is the introduction of **AI-assisted scheduling**. In the current implementation, AI modifies scheduler behaviour through deterministic optimisation rules that improve resource utilization while maintaining system stability. These adjustments may influence scheduling weights, resource allocation efficiency, and user prioritisation based on the selected AI mode.

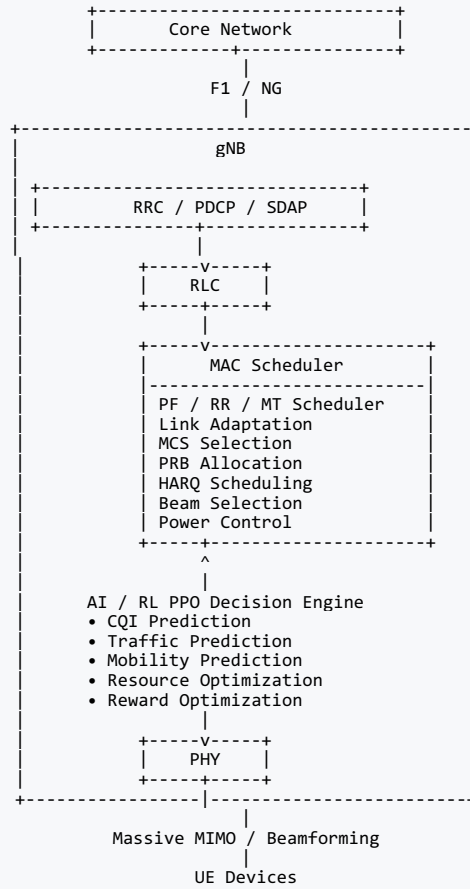
Future releases will replace these rule-based adjustments with **Reinforcement Learning (RL)**, where an AI agent continuously learns scheduling policies from network observations. Rather than relying on predefined optimisation rules, the RL scheduler will dynamically determine the most effective scheduling strategy by observing network state, selecting scheduling actions, and receiving rewards based on improvements in throughput, fairness, latency, and spectrum utilization. This evolution aligns with emerging AI-native RAN concepts being investigated for 6G networks.

3.7.5 Integration with the Engineering Workflow

Within RANOpt AI, scheduler operation follows completion of the radio channel calculations. The scheduler therefore receives the effective SINR, CQI, MCS, available bandwidth, and user demand before allocating radio resources. The resulting resource allocation directly determines the achievable throughput and KPI values displayed by the platform.

Figure 3.6 – MAC Scheduler within gNB

Figure 3.6 AI-Enhanced MAC Scheduler within the gNB

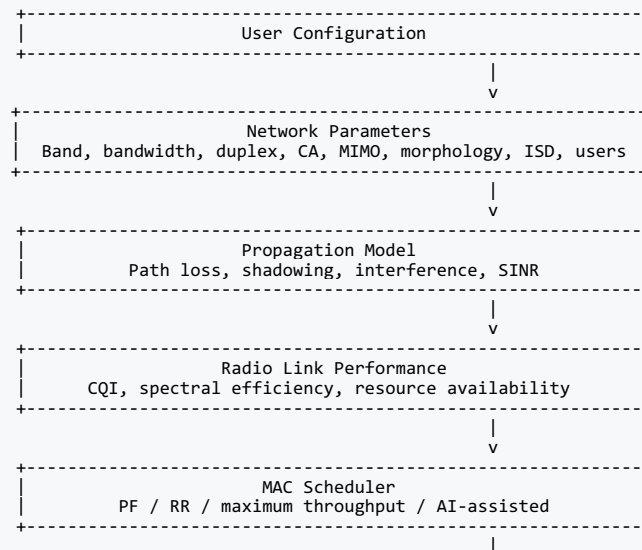


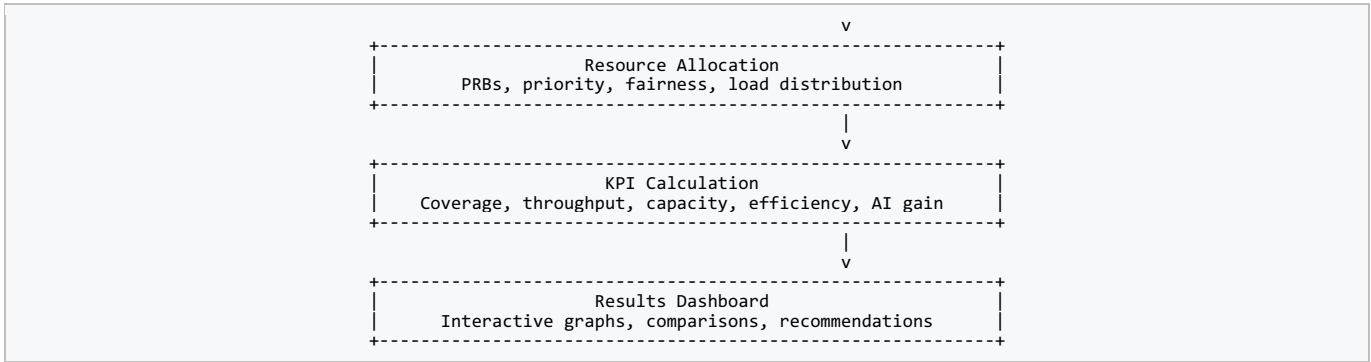
3.8 Integration of the MAC Scheduler into the RANOpt AI Workflow

The MAC scheduler forms the operational bridge between the radio environment, network configuration, and the AI optimization engine. Rather than operating as an isolated scheduling algorithm, it receives inputs from all major functional blocks of the simulator and converts these into practical resource allocation decisions that directly influence the network Key Performance Indicators (KPIs).

Figure 3.7 illustrates the overall workflow adopted in RANOpt AI.

Figure 3.7 – RANOpt AI Workflow with MAC Scheduler





Unlike a commercial base station implementation, the scheduler in RANOpt AI does not execute Transmission Time Interval (TTI)-level scheduling decisions. Instead, it represents the average behaviour of modern NR schedulers over many scheduling intervals. This abstraction significantly reduces computational complexity while preserving the relationships between scheduler behaviour and network-level performance.

The scheduler receives the estimated radio quality for each user together with the selected network configuration. Based on the selected scheduling strategy, it determines the relative allocation of available Physical Resource Blocks (PRBs) among active users. The resulting allocation determines achievable throughput, fairness, and overall cell utilisation.

The implementation currently supports several scheduler options commonly discussed in wireless systems:

- Round Robin scheduling for maximum fairness.
- Maximum Throughput scheduling for peak capacity.
- Proportional Fair scheduling providing a practical balance between fairness and capacity.
- AI-assisted scheduler employing reinforcement learning–based policy inference to optimise multiple KPIs simultaneously.

Because the scheduler lies at the centre of the processing chain, any improvement in scheduling decisions immediately affects nearly every performance metric presented by the simulator. Consequently, the scheduler serves as the principal point at which Artificial Intelligence is integrated into the RANOpt AI platform.

The browser implementation uses calibrated analytical models to approximate scheduler behaviour rather than reproducing every physical-layer procedure specified by 3GPP. This enables rapid interactive execution while maintaining realistic performance trends suitable for network planning, “what-if” analysis, and AI-assisted optimisation studies.

3.9 AI Integration with the Scheduler

Having established the baseline scheduler operation, the next stage introduces Artificial Intelligence into the decision-making process. Rather than replacing existing scheduling algorithms, the AI engine continuously evaluates the current network conditions and recommends scheduling policies that maximise overall network performance while maintaining fairness among users. The reinforcement learning framework employed in RANOpt AI is described in the following section.

This learning framework enables the scheduler to optimise long-term network performance rather than making isolated scheduling decisions.

3.9.1 Browser Implementation

The current browser-based implementation employs a practical approximation of the reinforcement learning framework. A fully trained RL agent normally requires extensive offline training using millions of scheduling iterations and realistic network datasets. Such training is computationally intensive and unsuitable for execution within a web browser.

This is exactly the kind of heavy, scriptable workload better suited to command-line and script-based tooling (see Section 1.4) than to an interactive browser session: offline PPO training typically runs as a batch job — using standard machine-learning frameworks driven from the command line — on a workstation, GPU server, or HPC cluster, well outside the browser environment.

Accordingly, RANOpt AI separates AI implementation into two stages.

Stage 1 – Offline Pre-training

The reinforcement learning model is trained offline using representative network scenarios spanning different morphologies, traffic conditions, spectrum allocations, user densities, and scheduler configurations. The resulting policy represents accumulated scheduling knowledge rather than a single simulation.

Training scenarios may be generated synthetically by the RANOpt AI engineering models across a broad sweep of representative conditions, or, where available, derived from anonymised carrier (operator) network data — such as traffic loading profiles, spectrum allocations, and scheduler configurations — exported from planning or operational systems into a common scenario format. Both sources feed the same offline training pipeline; the browser application itself never ingests raw carrier data, only the resulting trained policy. Full data preparation and export procedures are described in the companion AI document.

Stage 2 – Browser Inference

When the simulator is executed, the browser loads the pre-trained policy and performs inference only. For each selected scenario, the AI evaluates the current network state and applies the learned scheduling policy to estimate the resulting improvements in throughput, spectral efficiency, fairness, and other KPIs.

The trained policy itself is a compact set of model weights, exported from the offline training environment and packaged with the browser application so that it can be loaded at startup with no dependency on the original training data or infrastructure.

This architecture provides near-instantaneous response while avoiding the computational burden associated with real-time reinforcement learning training. It also ensures consistent and repeatable results across simulation sessions.

Future versions of RANOpt AI may optionally support incremental local retraining using operator-specific datasets, allowing the AI policy to evolve while preserving the interactive nature of the platform.

3.10 PPO Reinforcement Learning Training

Proximal Policy Optimisation (PPO) was selected for RANOpt AI because it provides stable, computationally efficient learning for sequential decision-making — well suited to wireless resource management, where conditions change continuously with traffic, mobility, interference, and propagation. Applied to MAC scheduling, PPO learns a policy that maximises long-term network performance rather than optimising a single scheduling interval. The general RL framework (state, action, reward, policy) is described in Section 2.7; full hyperparameters, training data, and deployment safeguards are in the companion AI document, RANOpt AI – AI Design & Implementation.

3.10.1 Reinforcement Learning Components

As in Section 2.7, the environment is the simulated 5G/6G radio access network; the state vector captures conditions such as cell loading, SINR/CQI, available PRBs, and scheduler configuration; actions adjust scheduling strategy, fairness weighting, and resource-allocation priorities; and the reward combines throughput, cell-edge throughput, spectral efficiency, fairness, and latency into a single training signal. The browser implementation represents these actions using calibrated analytical models rather than detailed TTI-level scheduling.

3.10.2 PPO Training Process

Training proceeds through repeated simulated scheduling episodes: the agent observes a network scenario, selects an action under its current policy, the simulated scheduler applies it, KPIs are recalculated, a reward is computed, and PPO updates the policy — repeated over many thousands of episodes until the policy converges toward actions that consistently maximise long-term performance.

3.10.3 Offline Pre-Training

As described in Section 3.9.1, training and inference are kept separate: PPO is trained offline across a wide range of morphologies, user densities, carrier configurations, and traffic conditions, and only the resulting trained policy is distributed with the browser application, which performs inference only. Future versions may support optional local policy refinement using operator-specific data, validated before deployment — the local-adaptation approach introduced here.

3.10.5 Advantages of PPO for RAN Optimization

PPO offers several advantages for RAN optimisation: stable, reliable convergence; efficient learning across large state spaces; robustness to changing network conditions; simultaneous optimisation of multiple KPIs; and good scalability to large numbers of users — advantages that have made it one of the most widely adopted RL algorithms for intelligent wireless resource management, and a strong foundation for RANOpt AI's evolution toward fully adaptive, AI-native 6G networks.

3.11 Chapter Summary

This chapter described the wireless models that support RANOpt AI, including propagation, link budget, SINR, channel abstraction, spectral efficiency, scheduling, and PPO-based decision support.

The propagation modelling adopted in RANOpt AI was intentionally designed as a system-level representation rather than a detailed physical-layer implementation. Large-scale propagation effects, including path loss, morphology, interference, and calibrated spectral efficiency models, are employed to estimate network performance while maintaining the computational efficiency required for interactive browser execution. Although the 3GPP TDL and CDL channel models provide greater physical-

layer fidelity, their full implementation is outside the scope of the current browser-based platform and represents a potential enhancement for future high-fidelity digital twin implementations.

The chapter then examined the role of the MAC scheduler as the central decision engine responsible for allocating radio resources among active users. Conventional scheduling algorithms such as Round Robin, Maximum Throughput, and Proportional Fair were reviewed, illustrating the trade-offs between throughput, fairness, and overall resource utilization.

The reinforcement learning implementation based on Proximal Policy Optimization (PPO) was then presented. The chapter described the definition of the environment, state space, action space, and reward function, together with the offline training methodology adopted by RANOpt AI. To achieve practical browser performance, reinforcement learning training is performed offline using representative network scenarios, while the browser executes only policy inference using the pre-trained model. This architecture provides fast execution, repeatable results, and realistic demonstrations of AI-assisted scheduling without requiring extensive computational resources during user interaction.

Overall, the architecture presented in this chapter demonstrates that practical AI-assisted radio resource optimization can be achieved within an interactive web-based platform by combining established wireless engineering models with modern reinforcement learning techniques. The resulting system provides operators, equipment vendors, researchers, and students with an intuitive environment for evaluating the impact of network configuration choices on key performance indicators while illustrating the growing role of AI in future 5G-Advanced and 6G radio access networks.

CHAPTER 4

Product Status, Validation, and Future Evolution

4.1 Present Product Definition

RANOpt AI is presently defined as an AI-assisted interactive 5G/6G RAN decision platform for engineering analysis, technology evaluation, and controlled what-if optimization. It combines standards-aligned radio engineering models, configurable network assumptions, deterministic calculations, AI-assisted recommendations, and graphical KPI comparison within a browser-based environment.

4.2 Engineering Value and Intended Use

The principal value of RANOpt AI is the ability to compare alternative radio configurations rapidly under a common set of assumptions. The engineer can first establish a baseline, modify one or more parameters, and then inspect the corresponding KPI changes. AI-assisted modes can be compared with conventional operation without changing the underlying scenario, making the effect of the optimization method easier to understand.

The platform is therefore best viewed as an engineering decision-support and demonstration environment. It can support early design studies, operator technology discussions, university and laboratory evaluation, algorithm comparison, and explanation of 5G-to-6G evolution. Final network decisions remain with the engineer and should be confirmed using operator data and established planning, laboratory, and field-validation processes.

4.3 Current Scope and Boundaries

RANOpt AI does not replace a commercial radio-planning suite, an operational support system, a self-organizing-network controller, or a detailed physical-layer and TTI-level simulator. The present calculations represent system-level engineering behaviour and are intended to reveal directional performance relationships rather than predict the exact performance of a specific deployed site.

Site-specific clutter, terrain, building geometry, antenna patterns, live interference, traffic traces, handover events, coverage holes, and operational failure statistics are outside the present implementation unless they are supplied through future data interfaces. Likewise, the browser performs rapid analytical evaluation and pretrained-policy inference; it is not intended to conduct large-scale PPO training during a normal user session.

These boundaries are deliberate. They preserve responsiveness, repeatability, explainability, and ease of deployment while providing a practical foundation for more detailed implementations.

4.4 Validation and Operator Evaluation

Product validation should proceed at several levels. Deterministic unit checks confirm equations, band and duplex constraints, noise figures, link-budget relationships, and scheduler behaviour. Golden engineering scenarios verify expected KPI direction under controlled parameter changes. Regression tests then confirm that software revisions do not unintentionally change previously validated results.

For operator evaluation, simulated outputs should be compared with representative planning-tool results, laboratory measurements, drive-test observations, and network counters where such data are available. Differences should be interpreted in light of model scope and input quality. This comparison process can also identify where calibration factors or additional models are required for a particular deployment.

A recommended adoption path is therefore to begin with transparent baseline scenarios, validate the direction and magnitude of selected KPI changes, and only then introduce operator-specific data or local policy adaptation under controlled conditions.

4.5 Evolution Toward an Area-Wide Digital Twin

The next major evolution is from a representative single-cell model to a multi-cell, area-wide engineering environment. Such an implementation would represent macro and small-cell locations, sector configurations, inter-cell interference, user distributions, traffic variation, and mobility across a selected geography. Each site or sector could retain an independently editable configuration while contributing to a common network-level KPI view.

Subsequent versions may incorporate geographic visualization, time-varying traffic, handover and mobility indicators, operator performance data, automated calibration, and constrained local learning. These capabilities would allow RANOpt AI to evolve into an AI-native digital-twin decision platform without abandoning the present principles of engineering control, transparency, and explainability.

The browser interface can remain the principal interaction layer, while computationally intensive simulation, policy training, and large data storage may be assigned to local or cloud-based services when the scale of the digital twin requires them.

4.6 Relationship to the Companion AI Document

This Product Description explains what RANOpt AI is, how its engineering workflow is organized, which radio models support its outputs, and how the platform can evolve. Detailed treatment of the Rule-Based Expert AI, PPO architecture, offline pretraining, reward design, local adaptation, explainability, and AI validation is provided separately in the companion document, RANOpt AI - AI Design & Implementation.

Separating the two documents keeps the product description focused while allowing the AI implementation strategy to be revised independently as algorithms, training data, and 6G standardization mature.

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CHAPTER 5

Validation, Testing and Engineering Results

5.1 Validation Philosophy

Validation is a fundamental element of RANOpt AI because the platform is intended to support engineering decisions rather than merely display calculated values. Every result must therefore be traceable to a defined configuration, a known engineering model, and a consistent sequence of calculations. Validation is applied both before execution, by checking the selected inputs, and after execution, by reviewing intermediate quantities and output KPIs for internal consistency.

The validation philosophy follows four complementary principles: configuration validity, calculation traceability, engineering plausibility, and repeatability. Configuration validity confirms that the selected band, duplex mode, bandwidth, carrier aggregation, antenna configuration, scheduler, and user assumptions form a permitted scenario. Calculation traceability confirms that each output can be related to the path-loss, link-budget, SINR, channel-quality, spectral-efficiency, and resource-allocation calculations that produced it. Engineering plausibility checks whether the resulting KPIs are within realistic ranges for the selected scenario. Repeatability ensures that the same deterministic inputs produce the same outputs.

Validation does not imply that the present single-cell analytical model reproduces every detail of a commercial planning tool or operational network. Instead, it confirms that the platform behaves consistently within its stated modelling scope and that directional comparisons remain technically meaningful. Site-specific terrain, clutter, live interference, traffic traces, handover statistics, and hardware implementation effects require separate field or operator validation when they are introduced.

Figure 5.1 – Layered validation workflow

5.2 Validation Engine

The Validation Engine is the automated consistency-checking layer of RANOpt AI. It operates alongside the Engineering Calculation Engine and examines both the selected inputs and the calculated outputs. Its purpose is to identify invalid combinations, unrealistic assumptions, numerical inconsistencies, and potential software defects before the results are used for engineering comparison.

The engine evaluates rules in several categories:

- Band and duplex consistency, including whether the selected NR band is normally associated with Frequency Division Duplex (FDD) or Time Division Duplex (TDD).
- Downlink and uplink bandwidth constraints, including independent bandwidth selection where FDD operation permits it and common-channel restrictions where they apply.
- Carrier aggregation validity, including the number of component carriers, total configured bandwidth, and compatibility of selected layers.
- MIMO and UE-capability consistency, ensuring that the assumed UE receive and transmit capability does not exceed the selected device class.
- Noise-figure, transmit-power, antenna-gain, and implementation-margin range checks.
- Propagation and morphology consistency, including reasonable frequency, inter-site distance, and cell-radius relationships.
- Scheduler parameter validity, including fairness coefficients, user-priority settings, and resource-allocation limits.
- AI configuration consistency, ensuring that AI-related comparisons are labelled correctly and that an AI-disabled baseline remains available.
- KPI range and monotonicity checks, such as non-negative throughput, coverage bounded by 0 and 100 percent, and resource utilization bounded by its defined range.

Each rule generates a pass, warning, or fail status. A pass indicates that the condition is consistent with the model. A warning identifies a technically possible but unusual or potentially misleading configuration that requires engineering review. A fail indicates an invalid combination or a calculation error that should be corrected before the result is accepted. This graded approach avoids rejecting legitimate expert studies while still protecting less experienced users from accidental misconfiguration.

5.3 Trace Mode

Trace Mode provides a step-by-step engineering view of the complete calculation chain. It is primarily intended for Expert Mode, debugging, model verification, and technical review. Instead of presenting only final KPIs, Trace Mode exposes the important intermediate values that connect user inputs to the displayed outputs.

Representative trace entries include:

- Selected frequency band, center frequency, duplex mode, and configured downlink and uplink bandwidth.
- Component-carrier configuration and total aggregated bandwidth.
- gNB transmit power, antenna gain, feeder and implementation losses, UE and gNB noise figures, and receiver sensitivity assumptions.
- Morphology, inter-site distance, representative user distance, and calculated path loss.
- Received signal level, thermal noise, interference allowance, and resulting downlink and uplink SINR.
- CQI estimate, selected MCS region, channel-efficiency factor, and spectral-efficiency estimate.
- Available physical resource blocks, scheduler utilization, active-user loading, and allocated resource share.
- Calculated cell throughput, average UE throughput, 5th-percentile throughput, coverage estimate, and AI-related KPI changes.

Trace Mode also records the active calculation path. For example, it identifies whether the scenario uses FDD alone or an FDD-plus-TDD layered configuration, whether carrier aggregation is enabled, which scheduler is active, and whether rule-based or PPO-assisted decision logic contributes to the displayed recommendation. This improves software maintainability because a suspicious output can be traced to the exact stage at which it was created.

The trace is an engineering diagnostic record, not a substitute for the main dashboard. Normal Mode therefore presents only the principal KPIs, while Expert Mode makes the trace available to users who require detailed verification.

5.4 Quick Validation Test

The Quick Validation Test is designed for immediate use after a configuration is selected. It executes a compact set of high-value checks and provides a rapid indication of whether the scenario is ready for analysis. The test is intentionally fast so that it can be run repeatedly during interactive what-if studies.

The Quick Test examines the following essential conditions:

- At least one valid carrier and channel bandwidth are selected.
- Duplex mode and band configuration are compatible.
- Downlink and uplink bandwidth values are positive and within the supported menu range.
- UE and gNB antenna configurations are internally consistent.
- Active-user count and traffic load are greater than zero and within the supported analytical range.
- Transmit power, noise figure, antenna gain, and morphology inputs are numerically valid.
- Scheduler and AI selections are defined.
- Calculated path loss, SINR, spectral efficiency, throughput, and coverage are finite and physically plausible.

The result is displayed as a concise pass/fail summary with warnings where appropriate. A passing Quick Test indicates that the scenario is suitable for interactive comparison; it does not represent full acceptance of every engineering assumption. A failed test identifies the parameter requiring correction and prevents ambiguous results from being interpreted as valid network behaviour.

5.5 Detailed Validation Test

The Detailed Validation Test performs a broader engineering review of the configuration, calculation chain, and KPI relationships. It is intended for final scenario review, software release checks, expert studies, and preparation of results for external discussion.

The Detailed Test expands the Quick Test by verifying:

- Propagation-model selection and consistency with morphology and frequency.
- Link-budget arithmetic for downlink and uplink, including the correct use of UE noise figure in the downlink receiver calculation and gNB noise figure in the uplink receiver calculation.
- Noise-floor calculation from bandwidth, thermal noise density, receiver noise figure, and implementation margin.

- SINR-to-CQI and CQI-to-spectral-efficiency mapping.
- Channel-efficiency and implementation factors applied to the achievable spectral efficiency.
- Physical resource block availability for the selected bandwidth and numerology assumptions.
- Scheduler allocation, fairness behaviour, and conservation of available radio resources.
- Consistency between cell throughput, average UE throughput, active users, and resource utilization.
- Expected directional behaviour when bandwidth, MIMO order, user loading, carrier aggregation, or AI mode is changed.
- Presence of both baseline and optimized values when an AI improvement is displayed.

The Detailed Test should also include a small collection of golden scenarios. These scenarios use fixed, documented input combinations with expected output ranges. They are not intended to prove universal accuracy; instead, they protect against regression when software changes are introduced. A calculation that previously produced a stable result should not change materially unless the underlying engineering model has intentionally been revised.

5.6 Golden Engineering Scenarios

Golden engineering scenarios provide repeatable reference cases for regression testing and technical demonstration. At least five scenarios should be maintained because a single reference case cannot exercise the full range of duplex, bandwidth, loading, propagation, and scheduler behaviour represented by the platform.

- Coverage-limited low-band FDD scenario with moderate bandwidth, conventional MIMO, and a relatively large representative cell radius.
- Balanced mid-band FDD scenario with moderate user loading and a proportional-fair scheduler.
- Capacity-oriented mid-band TDD scenario with wide bandwidth, higher-order MIMO, and high active-user loading.
- FDD-plus-TDD layered scenario demonstrating the division between low-band coverage and mid-band capacity.
- Carrier-aggregation and AI-assisted scenario comparing baseline scheduling with optimized scheduling under congestion.

For each golden scenario, the document or software test record should store the complete input configuration, acceptable KPI ranges, expected warning conditions, and the direction of change produced by selected parameter modifications. For example, increasing bandwidth should generally increase cell capacity when the scenario is bandwidth-limited, whereas increasing active-user loading should reduce average UE throughput when the available resources remain fixed. These directional checks are often more robust than relying on a single exact output value.

5.7 Validation Report

The Validation Report consolidates the results of the configuration checks, Quick Test, Detailed Test, and selected trace information into a reviewable engineering record. It allows the user to understand not only whether the scenario passed, but also which assumptions were used and which warnings require attention.

A complete report should include:

- Scenario name, date, software version, and operating mode.
- Complete radio configuration, morphology, traffic, scheduler, and AI selections.
- Pass, warning, and fail counts for each validation category.
- Key intermediate quantities from the propagation, link-budget, SINR, CQI, and scheduler calculations.
- Principal output KPIs and AI-versus-baseline differences.
- Engineering warnings, limitations, and model-scope statements.
- Optional user notes describing the purpose of the study and the decision reached.

The report supports reproducibility because another engineer can reconstruct the scenario from the recorded inputs. It also reduces the risk of circulating an isolated chart without the assumptions necessary to interpret it correctly.

5.8 Engineering Output Dashboard

The Engineering Output Dashboard is the primary results interface of RANOpt AI. It combines summary indicators, comparative charts, AI recommendations, and validation status in one browser workspace. The dashboard is organized to support a natural engineering sequence: first determine whether the configuration is valid, then review the principal KPIs, then examine detailed distributions and comparisons, and finally decide whether to retain or modify the configuration.

The principal dashboard elements include:

- KPI summary cards for coverage, cell throughput, average downlink and uplink UE throughput, 5th-percentile throughput, spectral efficiency, and resource utilization.
- AI On versus AI Off comparison for supported KPIs, with the baseline and optimized values clearly distinguished.
- Coverage and capacity-versus-distance charts showing the effect of morphology, frequency, bandwidth, and antenna assumptions.
- UE throughput distributions and cell-edge indicators that expose performance variation hidden by average values.
- SINR and CQI distributions that connect radio conditions to MCS and throughput behaviour.
- Per-layer or per-carrier capacity contribution where FDD-plus-TDD operation or carrier aggregation is selected.
- Scheduler statistics, including fairness and utilization indicators.
- UE uplink-signalling reduction estimates when AI-assisted signalling optimization is enabled.
- Explainable AI recommendations, confidence or priority indicators, and expected KPI trade-offs.
- Validation status and direct access to Quick Test, Detailed Test, and Trace Mode.

Normal Mode emphasizes the principal summary cards and charts required for rapid evaluation. Expert Mode adds detailed radio parameters, trace information, distributions, scheduler diagnostics, validation results, and advanced comparisons. Both modes use the same underlying calculation engine; the difference lies in the amount of information exposed to the user.

5.9 Engineering Charts and KPI Interpretation

Charts are included to reveal relationships that cannot be understood from a single numerical KPI. The engineering value of each chart depends on correct interpretation and on recognition of the model boundaries.

Coverage charts illustrate the relationship among path loss, received signal level, SINR threshold, and representative user distance. They should be interpreted as analytical coverage indicators for the selected morphology rather than as geographic predictions of streets, buildings, terrain, or individual coverage holes.

Throughput-versus-distance charts show the progressive reduction in achievable rate as radio conditions deteriorate. The trend reflects SINR, channel quality, spectral efficiency, available bandwidth, MIMO configuration, and scheduler resource sharing. A smooth analytical curve is expected because the current implementation does not reproduce every short-term fading event or discrete field measurement.

UE throughput-distribution charts provide a statistical view of user performance. Average throughput alone may conceal poor cell-edge behaviour; consequently, the 5th-percentile or cell-edge result is displayed as a separate KPI. Scheduler comparisons should examine both overall cell throughput and distribution fairness because an aggressive maximum-throughput strategy may increase the total rate while reducing service to disadvantaged users.

AI-improvement charts must always retain the non-AI baseline. Percentage improvement without the original value can be misleading, especially when the baseline is very small. RANOpt AI therefore presents the baseline, optimized value, and percentage change together wherever practical.

5.10 Comparative AI and Non-AI Analysis

AI-assisted performance is evaluated by running the same engineering scenario with an unchanged set of radio assumptions and comparing the baseline scheduler or configuration with the AI-recommended alternative. This controlled comparison is necessary because an apparent improvement should not be attributed to AI when it was actually produced by a simultaneous change in bandwidth, spectrum, MIMO, traffic, or propagation assumptions.

The comparison may include cell throughput, average UE throughput, 5th-percentile throughput, spectral efficiency, fairness, resource utilization, congestion indicators, and UE signalling reduction. The AI result is considered useful only when the improvement is technically plausible and does not create an unacceptable deterioration in another KPI.

Rule-based AI recommendations are directly explainable because the triggering engineering condition is known. PPO-based recommendations require an additional explanation layer that identifies the important state variables, selected action, expected reward contribution, and predicted trade-offs. In both cases, the engineer retains the authority to accept, reject, or modify the recommendation.

Figure 5.2 – Controlled AI-versus-baseline comparison

5.11 Engineering Workflow for Results Review

A disciplined results-review workflow prevents isolated KPI values from being accepted without understanding the assumptions that produced them. The recommended workflow is:

- Define the engineering question and establish the baseline scenario.

- Run the Quick Validation Test and correct any failed configuration checks.
- Execute the baseline calculation with AI disabled.
- Review coverage, throughput, spectral efficiency, loading, and cell-edge performance.
- Enable AI or select an alternative scheduler while holding unrelated parameters constant.
- Compare baseline and optimized KPIs, including adverse trade-offs.
- Use Trace Mode and the Detailed Validation Test for unexpected results.
- Save or export the validated scenario and record the engineering decision.

This workflow makes RANOpt AI an interactive decision platform rather than a one-button optimizer. The purpose is not simply to produce the highest number, but to help the engineer understand why performance changes and whether the change is suitable for the intended network objective.

5.12 Export and Reporting

Export and reporting functions allow validated engineering results to be retained outside the active browser session. A reportable scenario should include both the input configuration and the output KPIs so that the result remains understandable after it has been separated from the application.

Planned or supported export formats may include printable browser reports, PDF summaries, CSV tables for further analysis, and structured scenario files for reloading a configuration. Where charts are exported, the associated assumptions, software version, AI mode, and validation status should accompany the image or report.

Exported results should be treated as engineering study outputs rather than certified operational predictions. Final deployment decisions continue to require the operator's normal planning, laboratory, field, and acceptance procedures.

5.13 Limitations of Validation

The validation functions confirm consistency within the implemented model; they cannot validate information that the model does not contain. The current platform does not directly verify site-specific terrain and clutter, building penetration, antenna azimuth and tilt patterns, neighbour-list behaviour, handover failures, packet-level protocol timing, live interference, or operational hardware impairments. These aspects require richer multi-cell modelling or comparison with external planning and network data.

Similarly, a successful software test does not establish that an AI policy is safe for autonomous live-network control. The present product is an advisory engineering tool. Any future closed-loop implementation would require additional policy constraints, out-of-distribution detection, rollback procedures, security controls, operator approval rules, and extensive laboratory and field validation.

5.14 Chapter Summary

This chapter described the validation, testing, diagnostic, visualization, and reporting capabilities that support trustworthy use of RANOpt AI. The Validation Engine checks the compatibility of user selections and the internal consistency of calculated results. Quick Validation provides an immediate readiness check, while Detailed Validation and golden engineering scenarios support deeper technical review and software regression testing.

Trace Mode exposes the principal stages of the calculation chain, enabling engineers to relate displayed KPIs to the selected radio configuration, propagation assumptions, link budget, SINR, channel quality, spectral efficiency, scheduler behaviour, and resource allocation. The Validation Report preserves these assumptions and results in a reproducible record.

The Engineering Output Dashboard then converts the validated calculations into KPI summaries, distributions, coverage and throughput charts, scheduler indicators, AI recommendations, and controlled AI-versus-baseline comparisons. Normal and Expert modes provide different levels of detail while using the same underlying engineering calculations.

Together, these capabilities reinforce the central RANOpt AI philosophy: the platform performs transparent calculations and provides explainable recommendations, but the wireless engineer verifies the evidence, evaluates the trade-offs, and makes the final decision.